Nebico Oil Retention Project

Coolmay L10S Ladder program
TK8050 x 3, TK8100 x 1

Modifications:

1. Pointer Functions to shorten bulk calculation
2. Revised Core logic (Mode Scheduler)
3. Revised Oil Distribution
4. Clean Segmentation

Prepared By: Sambridda Ranabhat (Architect & Programmer)

09/09/2026
Mechatronics Engineering .pvt .ltd

Data Name : Memory Capacity
Memory Capacity

[Memory Capacity]
16000

[Program Capacity]
16000 Steps

[Comments Capacity]
0 Block 0 Points

[File Register Capacity]
0 Block 0 Points

[Special Function Block Setting]
0 Block

[Positioning]
0 Block

[Built-in CC-Link/LT Setting]
0 Block

Data Name : PLC System
PLC System

[Battery Less Mode]
OFF

[MODEM Initialized]
None

[RUN Terminal Input]
None

[Device]

	Sym.	Dig.	Points	Start	End	Latch Start	End	Latch Setting Range
Supplemental Relay	M	10	7680	0	7679	500	1023	0 - 1023
State	S	10	4096	0	4095	500	999	0 - 999
Timer	T	10	512	0	511			
Counter(16bit)	C	10	200	0	199	100	199	0 - 199
Counter(32bit)	C	10	56	200	255	220	255	200 - 255
Data Register	D	10	8000	0	7999	200	511	0 - 511
Extended Register	R	10	32768	0	32767			

CC-Link Setting

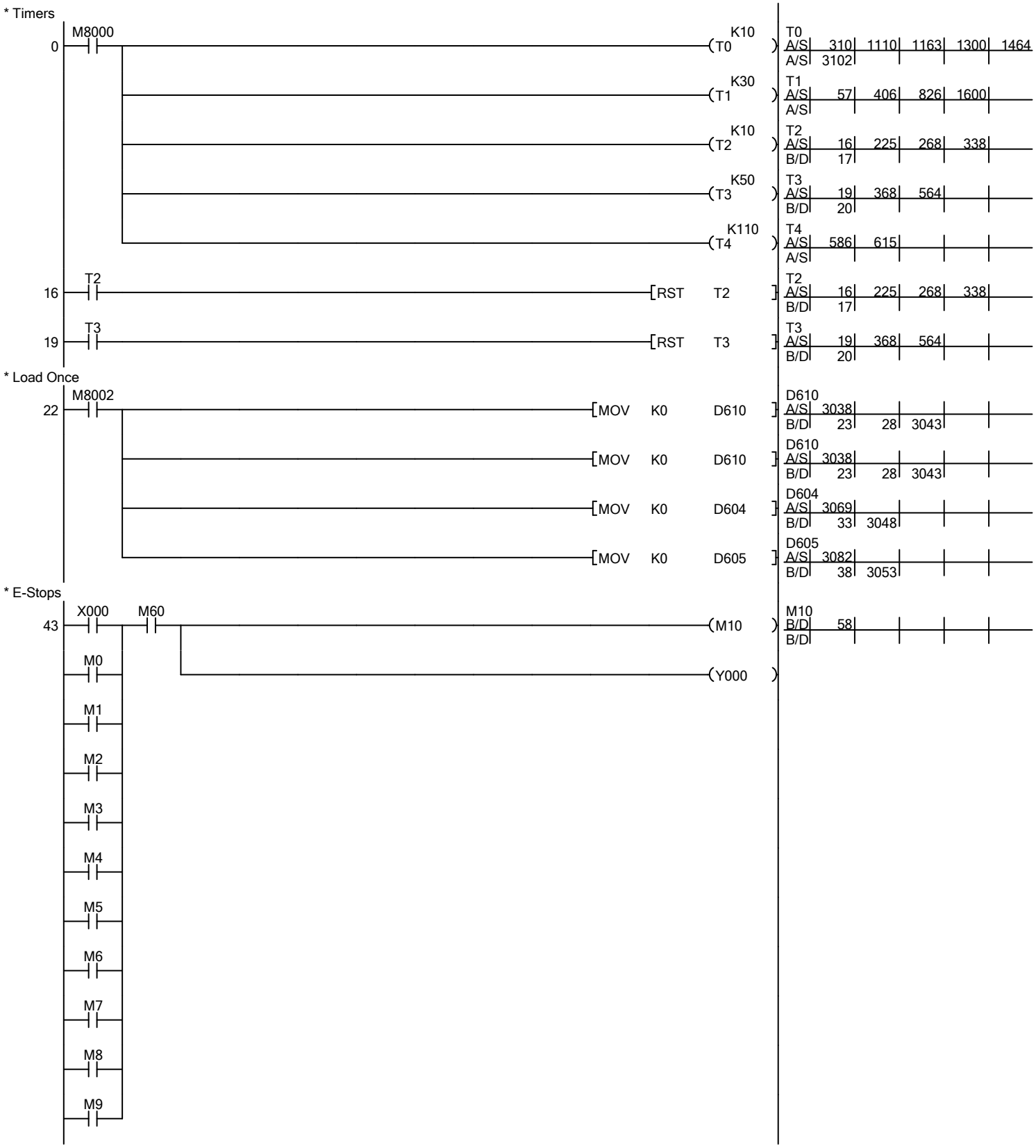
[CC-Link Setting]

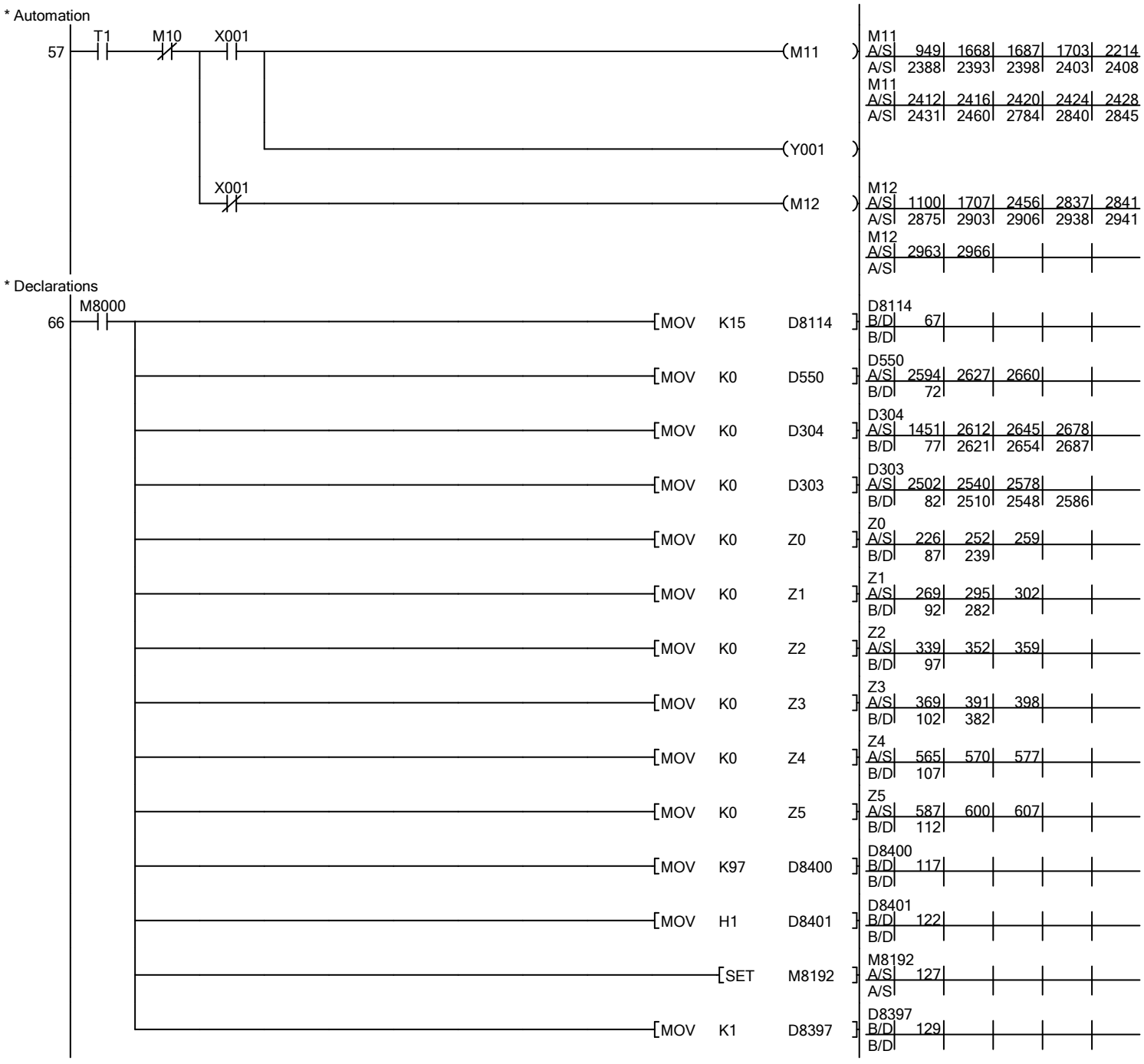
Connection Block Not Set

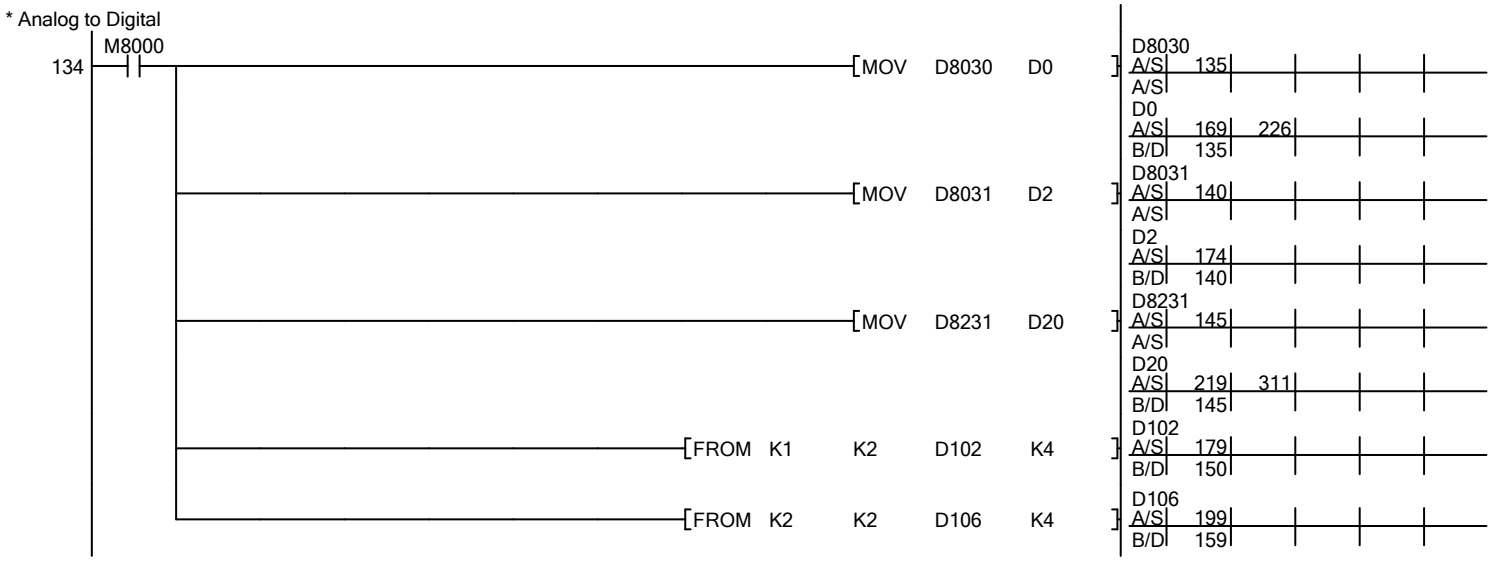
Data Name : CC-Link
CC-Link Setting

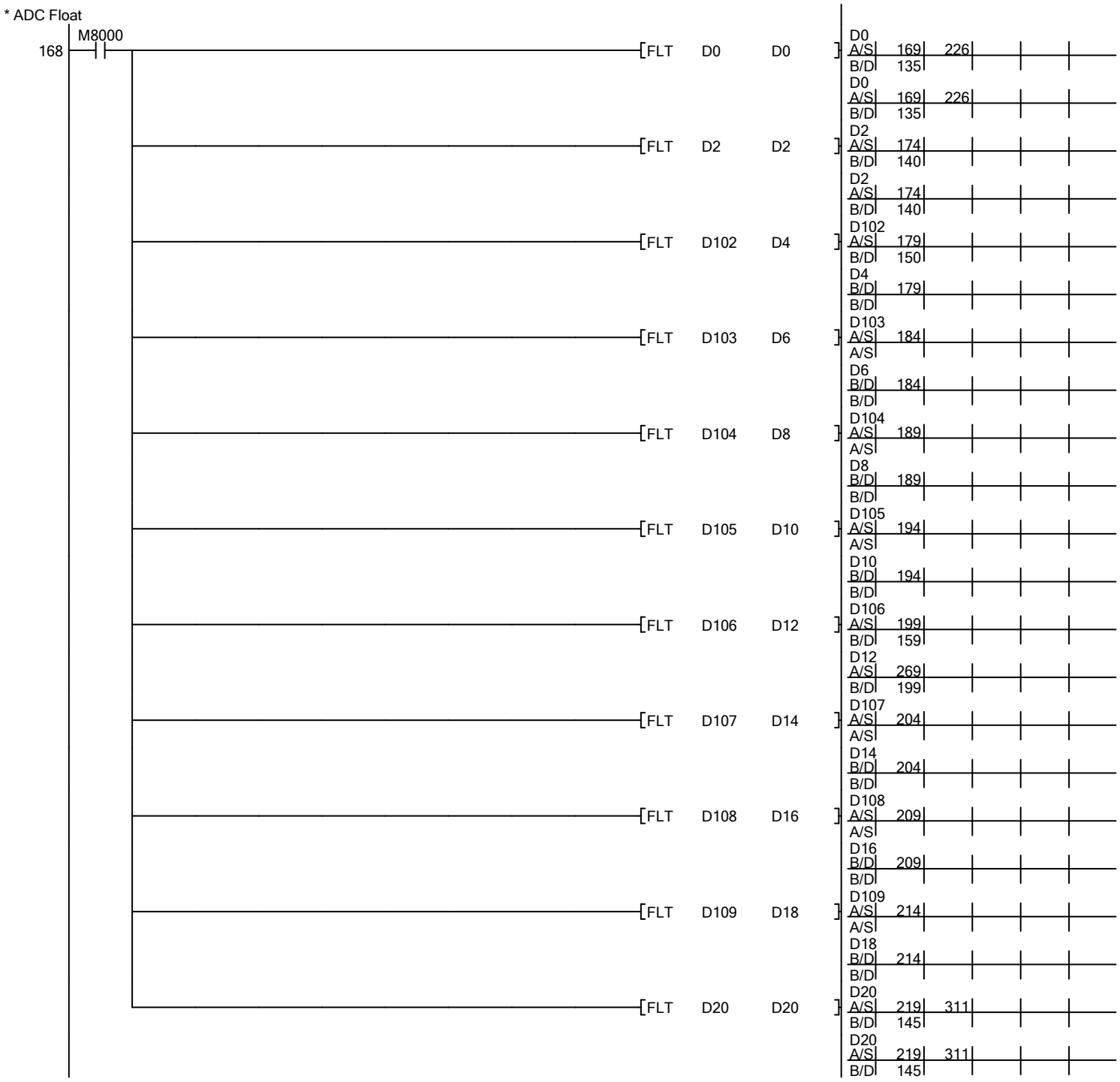
[CC-Link Setting]

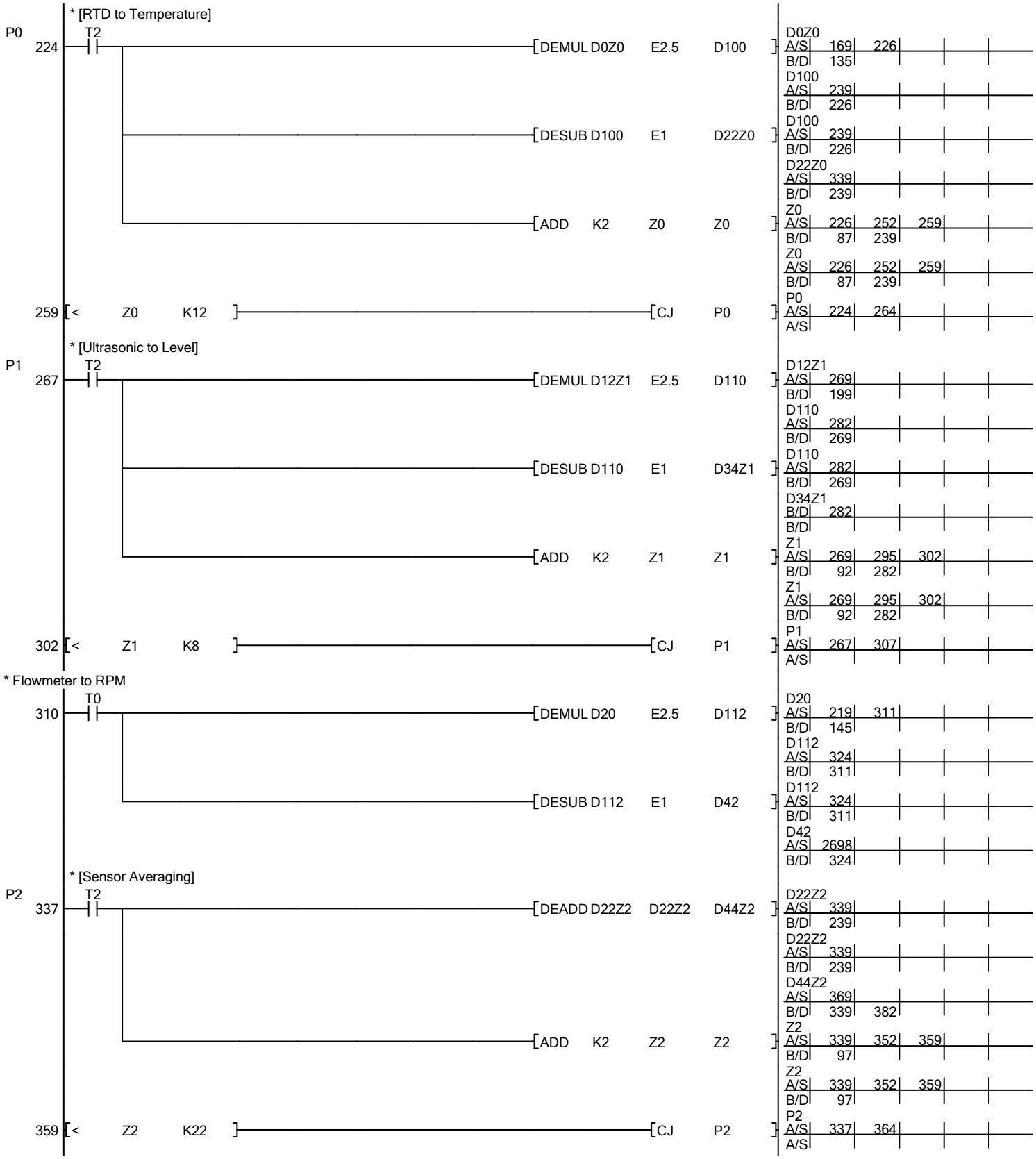
	Setting Contents
Special Function Block No.	-
Type	-
Master Station Data Link Type	Start Parameter by BFM
Mode	-
Total Module Connected	-
Remote Input(RX)	-
Remote Output(RY)	-
Remote Register(RWr)	-
Remote Register(RWw)	-
Ver.2 Remote Input(RX)	-
Ver.2 Remote Output(RY)	-
Ver.2 Remote Register(RWr)	-
Ver.2 Remote Register(RWw)	-
Special Relay(SB)	-
Special Register(SW)	-
Retry Count	-
Automatic Reconnection Station Count	-
Standby Master Station No.	-
PLC Down Select	-
Scan Mode Setting	-
Delay Time Setting	-
Remote Device Station Initial Setting	-
Interrupt Settings	-

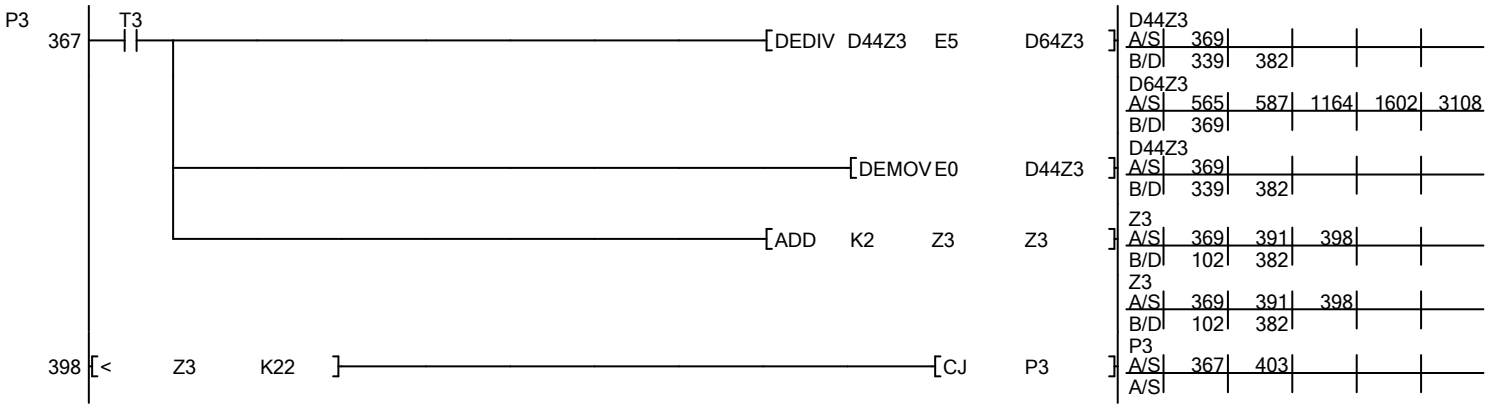

* Timers
0
M8000
K10
T0
T0
A/S 310 1110 1163 1300 1464
A/S 3102
K30
T1
T1
A/S 57 406 826 1600
A/S
K10
T2
T2
A/S 16 225 268 338
B/D 17
K50
T3
T3
A/S 19 368 564
B/D 20
K110
T4
T4
A/S 586 615
A/S
16
T2
RST T2
T2
A/S 16 225 268 338
B/D 17
19
T3
RST T3
T3
A/S 19 368 564
B/D 20
* Load Once
22
M8002
MOV K0 D610
D610
A/S 3038
B/D 23 28 3043
MOV K0 D610
D610
A/S 3038
B/D 23 28 3043
MOV K0 D604
D604
A/S 3069
B/D 33 3048
MOV K0 D605
D605
A/S 3082
B/D 38 3053
* E-Stops
43
X000
M60
M10
M10
B/D 58
B/D
Y000
M0
M1
M2
M3
M4
M5
M6
M7
M8
M9

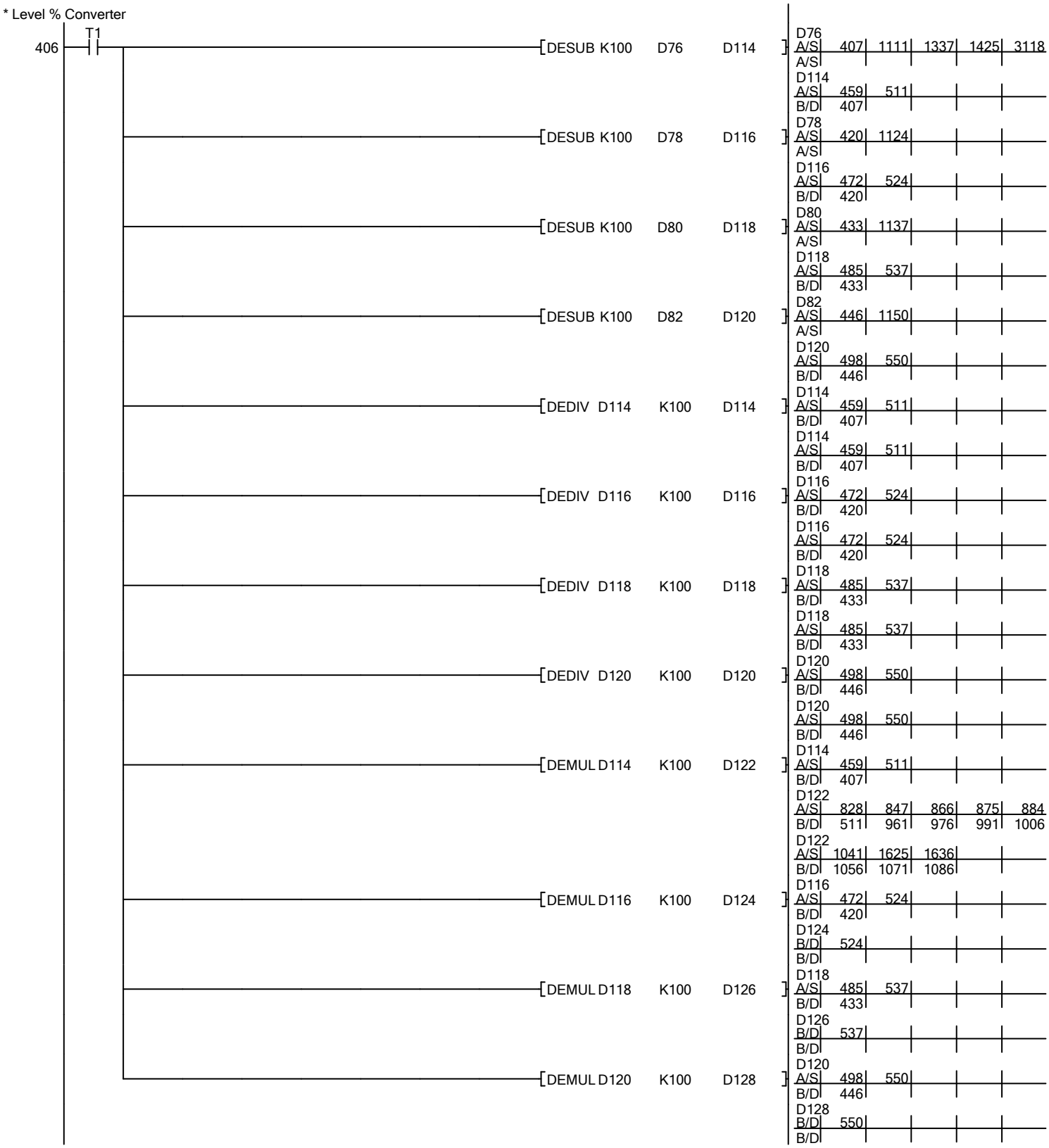

* Automation
57
T1
M10
X001
M11
M11
A/S 949 1668 1687 1703 2214
A/S 2388 2393 2398 2403 2408
M11
A/S 2412 2416 2420 2424 2428
A/S 2431 2460 2784 2840 2845
Y001
X001
M12
M12
A/S 1100 1707 2456 2837 2841
A/S 2875 2903 2906 2938 2941
M12
A/S 2963 2966
A/S
* Declarations
66
M8000
MOV K15 D8114
D8114
B/D 67
B/D
MOV K0 D550
D550
A/S 2594 2627 2660
B/D 72
MOV K0 D304
D304
A/S 1451 2612 2645 2678
B/D 77 2621 2654 2687
MOV K0 D303
D303
A/S 2502 2540 2578
B/D 82 2510 2548 2586
MOV K0 Z0
Z0
A/S 226 252 259
B/D 87 239
MOV K0 Z1
Z1
A/S 269 295 302
B/D 92 282
MOV K0 Z2
Z2
A/S 339 352 359
B/D 97
MOV K0 Z3
Z3
A/S 369 391 398
B/D 102 382
MOV K0 Z4
Z4
A/S 565 570 577
B/D 107
MOV K0 Z5
Z5
A/S 587 600 607
B/D 112
MOV K97 D8400
D8400
B/D 117
B/D
MOV H1 D8401
D8401
B/D 122
B/D
SET M8192
M8192
A/S 127
A/S
MOV K1 D8397
D8397
B/D 129
B/D

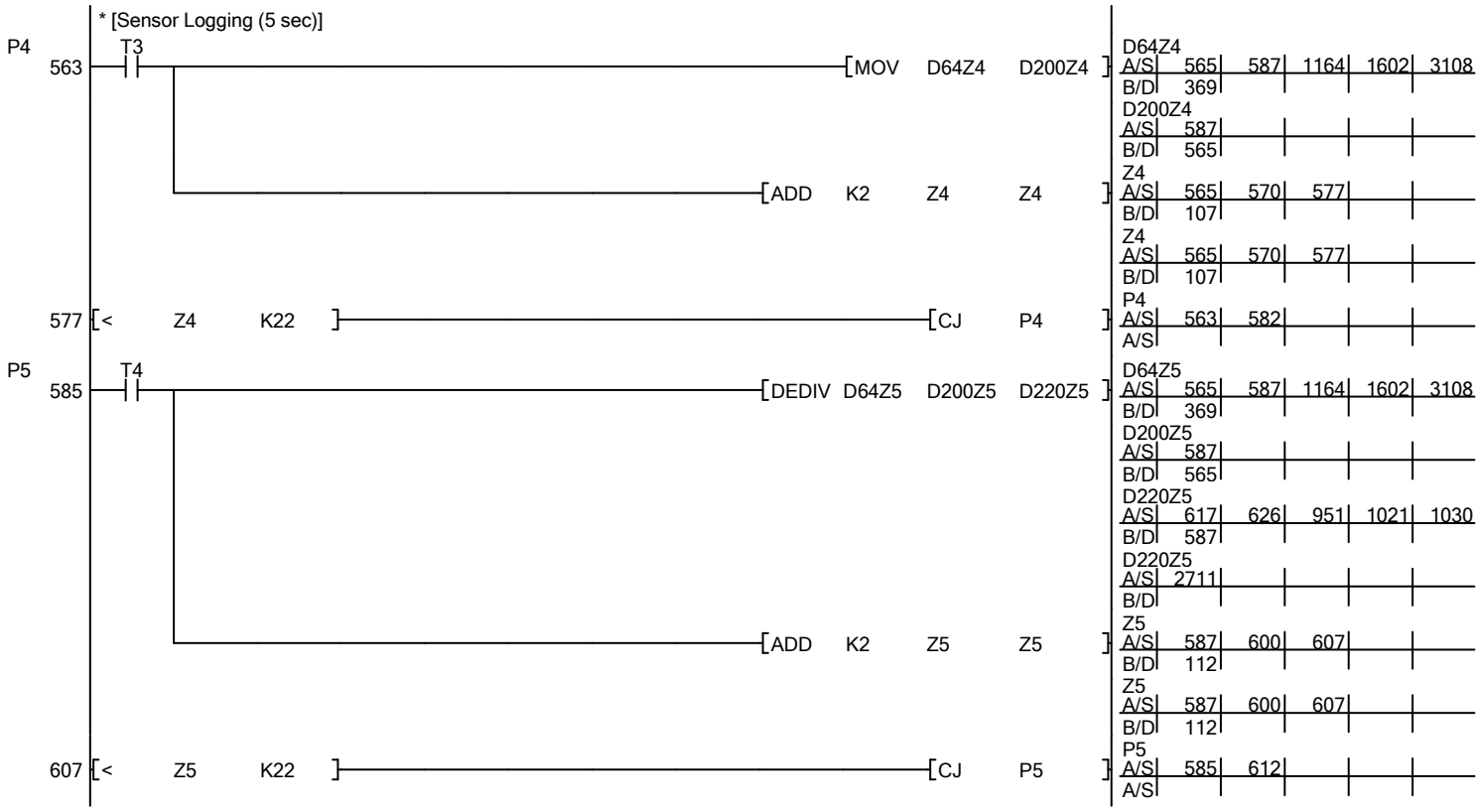

* Analog to Digital
M8000
134
MOV D8030 D0
MOV D8031 D2
MOV D8231 D20
FROM K1 K2 D102 K4
FROM K2 K2 D106 K4
D8030
A/S 135
A/S
D0
A/S 169 226
B/D 135
D8031
A/S 140
A/S
D2
A/S 174
B/D 140
D8231
A/S 145
A/S
D20
A/S 219 311
B/D 145
D102
A/S 179
B/D 150
D106
A/S 199
B/D 159

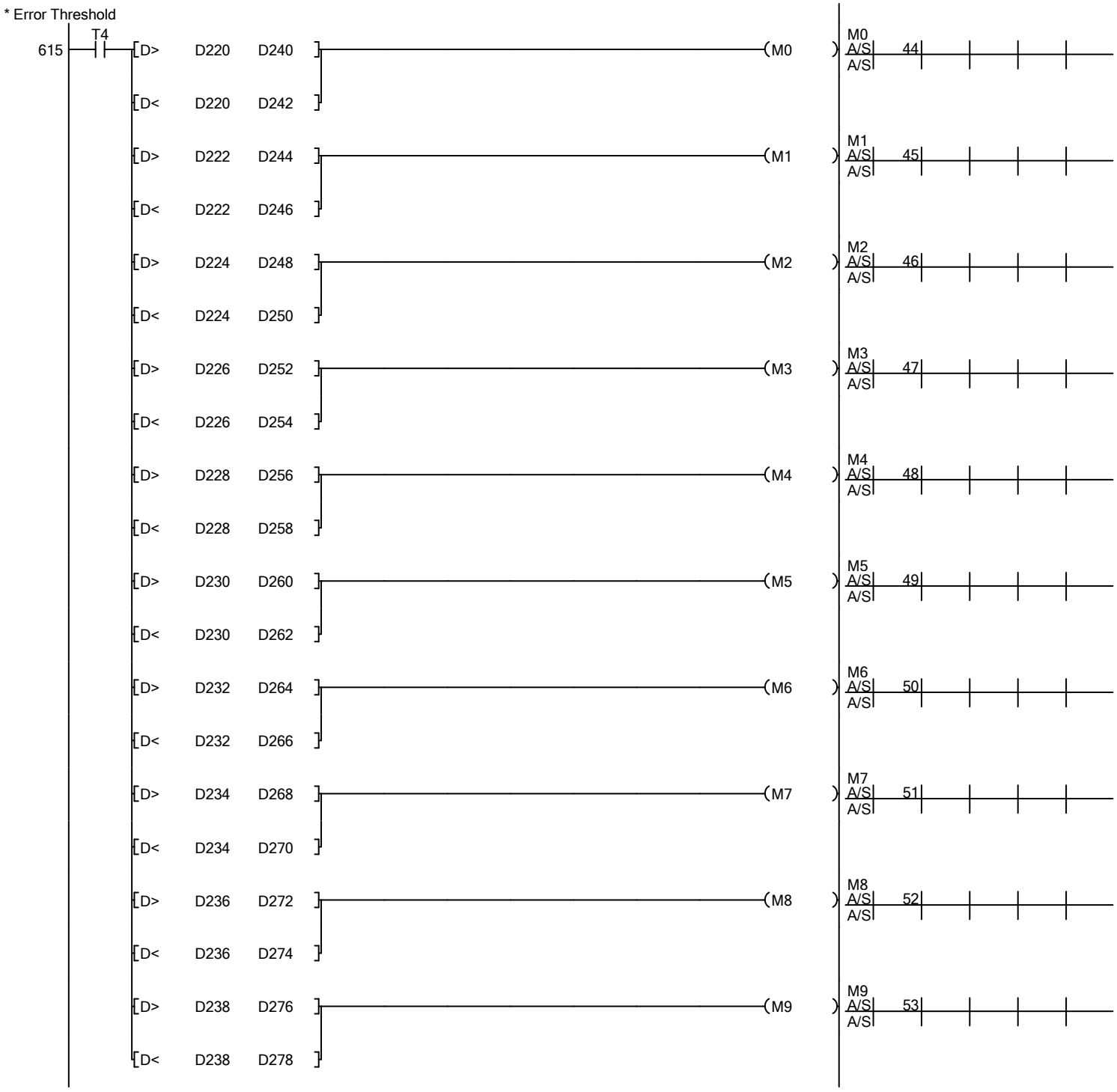

* ADC Float
M8000
168
FLT D0 D0
FLT D2 D2
FLT D102 D4
FLT D103 D6
FLT D104 D8
FLT D105 D10
FLT D106 D12
FLT D107 D14
FLT D108 D16
FLT D109 D18
FLT D20 D20
D0 A/S 169 226
B/D 135
D0 A/S 169 226
B/D 135
D2 A/S 174
B/D 140
D2 A/S 174
B/D 140
D102 A/S 179
B/D 150
D4 B/D 179
B/D
D103 A/S 184
A/S
D6 B/D 184
B/D
D104 A/S 189
A/S
D8 B/D 189
B/D
D105 A/S 194
A/S
D10 B/D 194
B/D
D106 A/S 199
B/D 159
D12 A/S 269
B/D 199
D107 A/S 204
A/S
D14 B/D 204
B/D
D108 A/S 209
A/S
D16 B/D 209
B/D
D109 A/S 214
A/S
D18 B/D 214
B/D
D20 A/S 219 311
B/D 145
D20 A/S 219 311
B/D 145


* [RTD to Temperature]
P0
T2
224
DEMUL D0Z0 E2.5 D100
DESUB D100 E1 D22Z0
ADD K2 Z0 Z0
259 < Z0 K12
CJ P0
* [Ultrasonic to Level]
P1
T2
267
DEMUL D12Z1 E2.5 D110
DESUB D110 E1 D34Z1
ADD K2 Z1 Z1
302 < Z1 K8
CJ P1
* Flowmeter to RPM
T0
310
DEMUL D20 E2.5 D112
DESUB D112 E1 D42
* [Sensor Averaging]
P2
T2
337
DEADD D22Z2 D22Z2 D44Z2
ADD K2 Z2 Z2
359 < Z2 K22
CJ P2

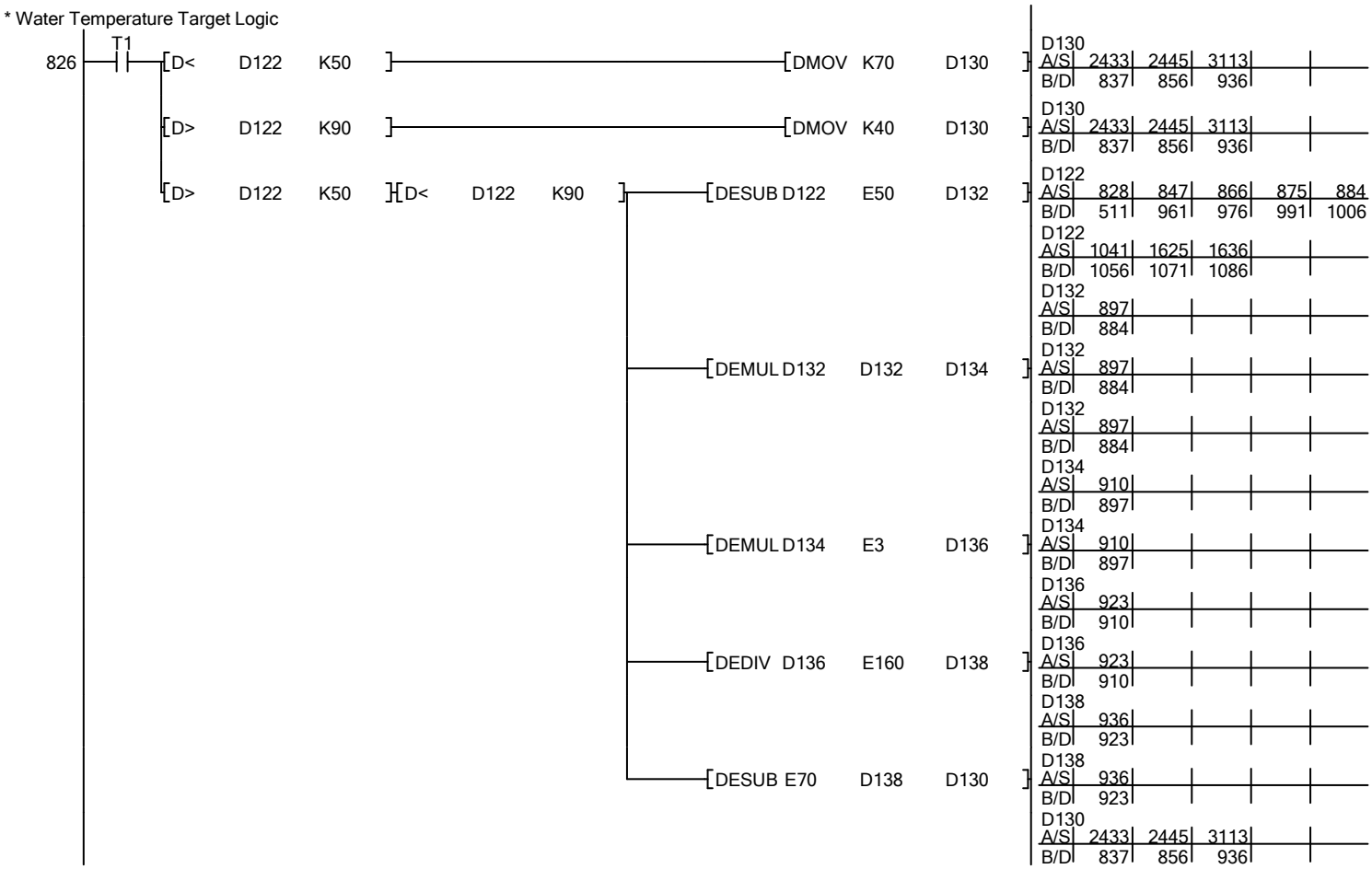

P3
T3
367
DEDIV D44Z3 E5 D64Z3
DEMOV E0 D44Z3
ADD K2 Z3 Z3
398
< Z3 K22
CJ P3
D44Z3
A/S 369
B/D 339 382
D64Z3
A/S 565 587 1164 1602 3108
B/D 369
D44Z3
A/S 369
B/D 339 382
Z3
A/S 369 391 398
B/D 102 382
Z3
A/S 369 391 398
B/D 102 382
P3
A/S 367 403
A/S

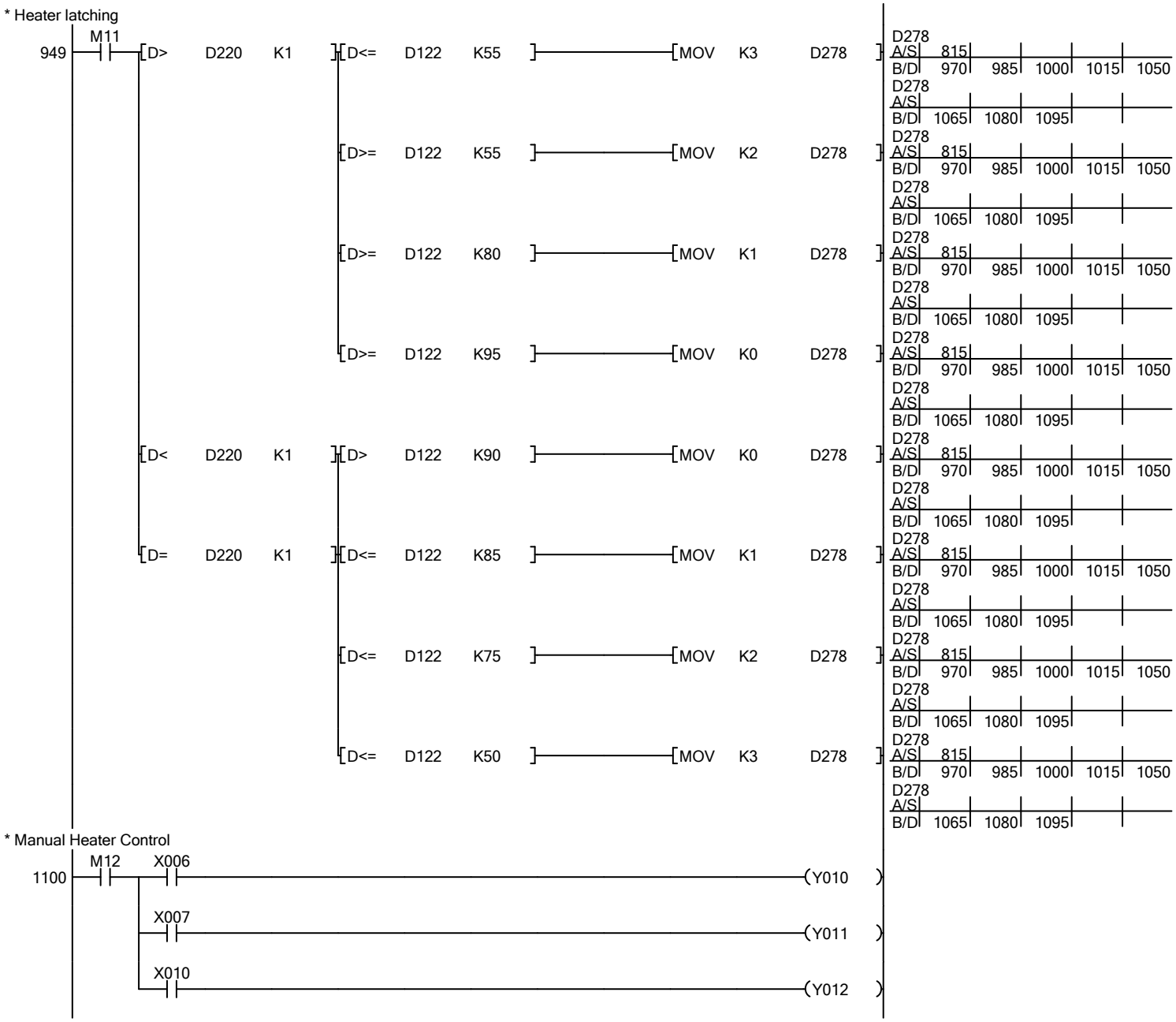

* Level % Converter
406
T1
[DESUB K100 D76 D114]
[DESUB K100 D78 D116]
[DESUB K100 D80 D118]
[DESUB K100 D82 D120]
[DEDIV D114 K100 D114]
[DEDIV D116 K100 D116]
[DEDIV D118 K100 D118]
[DEDIV D120 K100 D120]
[DEMUL D114 K100 D122]
[DEMUL D116 K100 D124]
[DEMUL D118 K100 D126]
[DEMUL D120 K100 D128]
D76 A/S 407 1111 1337 1425 3118
A/S
D114 A/S 459 511
B/D 407
D78 A/S 420 1124
A/S
D116 A/S 472 524
B/D 420
D80 A/S 433 1137
A/S
D118 A/S 485 537
B/D 433
D82 A/S 446 1150
A/S
D120 A/S 498 550
B/D 446
D114 A/S 459 511
B/D 407
D114 A/S 459 511
B/D 407
D116 A/S 472 524
B/D 420
D116 A/S 472 524
B/D 420
D118 A/S 485 537
B/D 433
D118 A/S 485 537
B/D 433
D120 A/S 498 550
B/D 446
D120 A/S 498 550
B/D 446
D114 A/S 459 511
B/D 407
D122 A/S 828 847 866 875 884
B/D 511 961 976 991 1006
D122 A/S 1041 1625 1636
B/D 1056 1071 1086
D116 A/S 472 524
B/D 420
D124 B/D 524
B/D
D118 A/S 485 537
B/D 433
D126 B/D 537
B/D
D120 A/S 498 550
B/D 446
D128 B/D 550
B/D

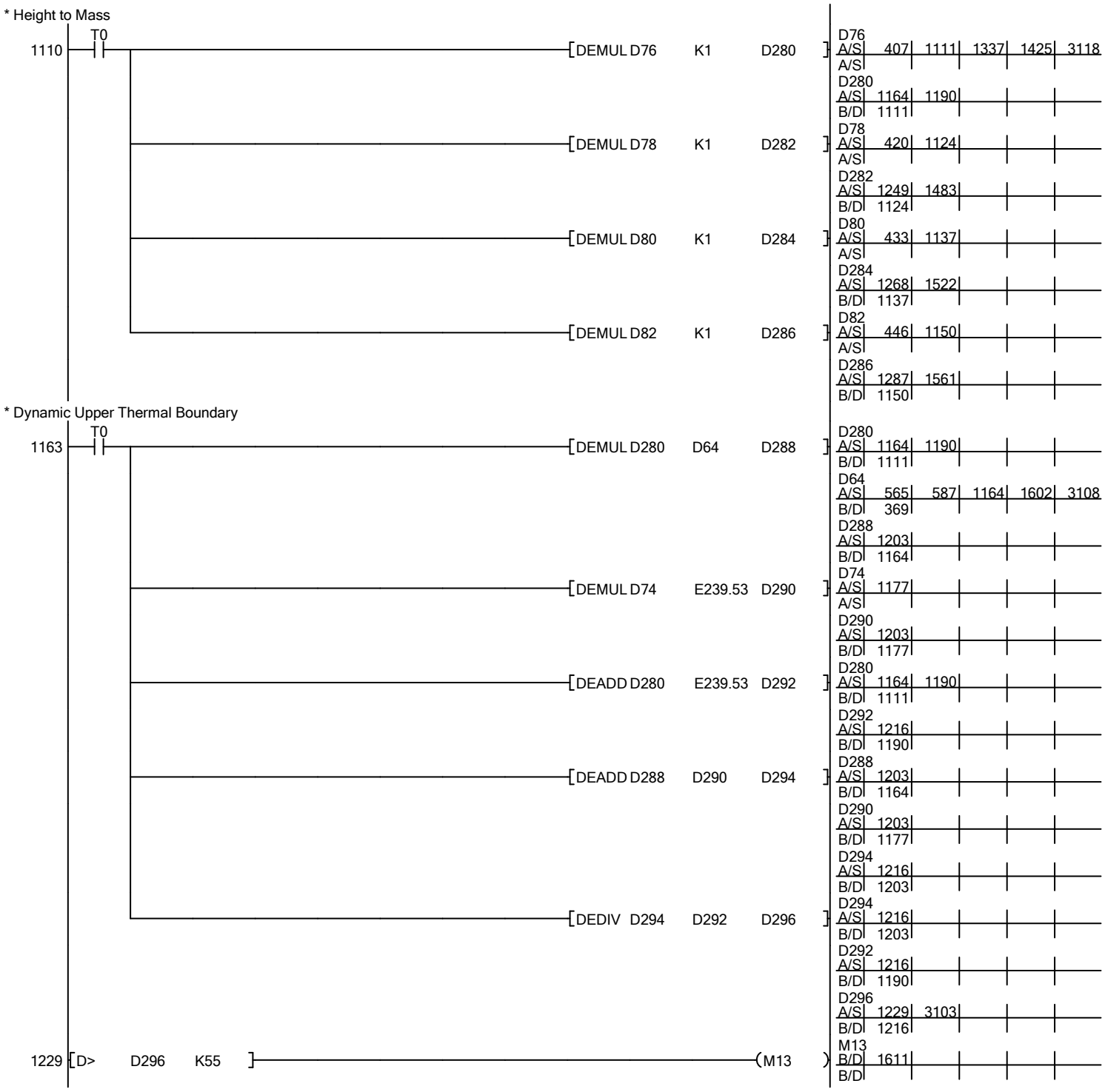

* [Sensor Logging (5 sec)]
P4
563
T3
MOV D64Z4 D200Z4
ADD K2 Z4 Z4
577
< Z4 K22
CJ P4
P5
585
T4
DEDIV D64Z5 D200Z5 D220Z5
ADD K2 Z5 Z5
607
< Z5 K22
CJ P5
D64Z4 A/S 565 587 1164 1602 3108 B/D 369
D200Z4 A/S 587 B/D 565
Z4 A/S 565 570 577 B/D 107
Z4 A/S 565 570 577 B/D 107
P4 A/S 563 582 A/S
D64Z5 A/S 565 587 1164 1602 3108 B/D 369
D200Z5 A/S 587 B/D 565
D220Z5 A/S 617 626 951 1021 1030 B/D 587
D220Z5 A/S 2711 B/D
Z5 A/S 587 600 607 B/D 112
Z5 A/S 587 600 607 B/D 112
P5 A/S 585 612 A/S

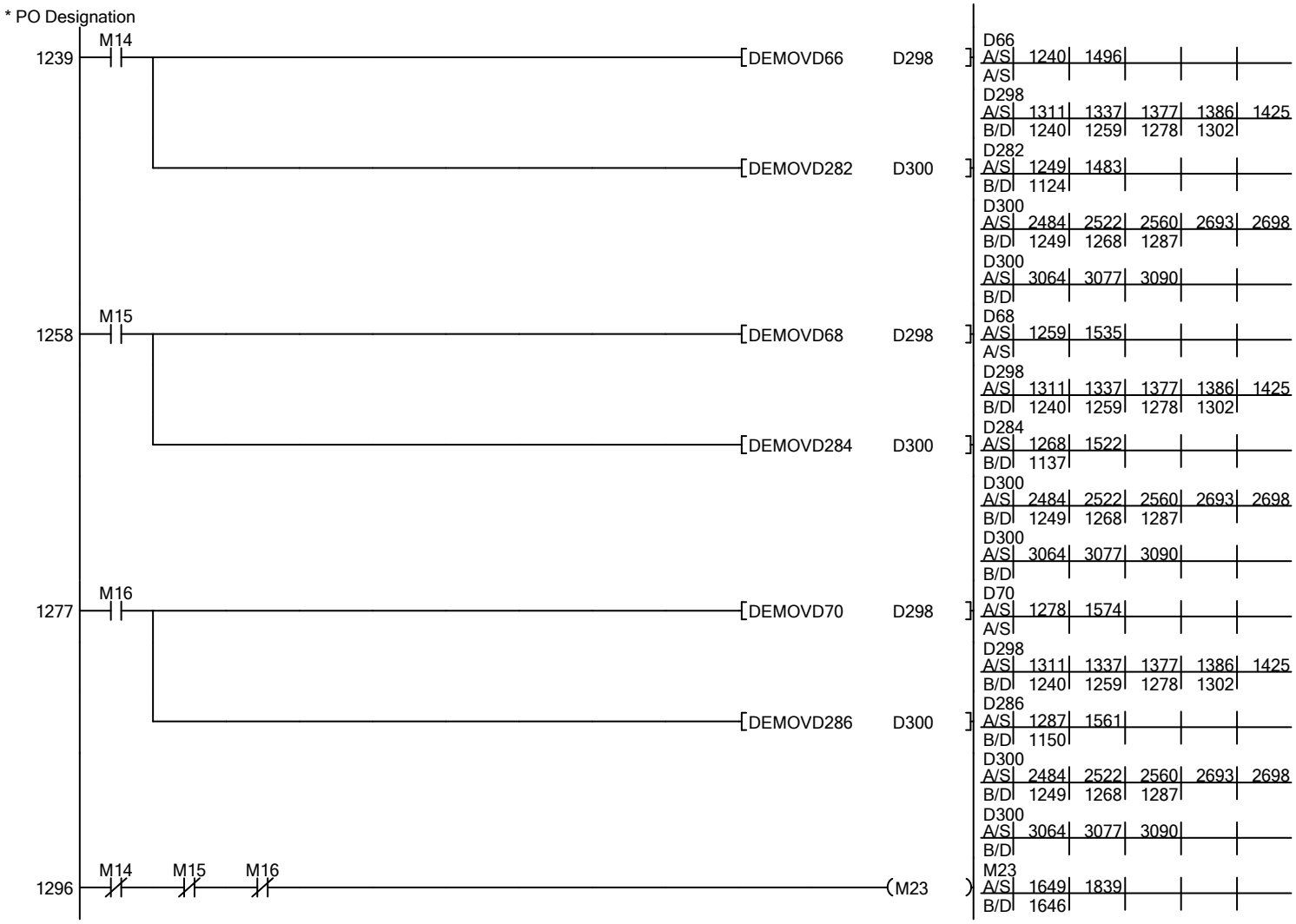

* Error Threshold
615
T4
D> D220 D240
D< D220 D242
D> D222 D244
D< D222 D246
D> D224 D248
D< D224 D250
D> D226 D252
D< D226 D254
D> D228 D256
D< D228 D258
D> D230 D260
D< D230 D262
D> D232 D264
D< D232 D266
D> D234 D268
D< D234 D270
D> D236 D272
D< D236 D274
D> D238 D276
D< D238 D278
M0
M1
M2
M3
M4
M5
M6
M7
M8
M9
M0 A/S 44 A/S
M1 A/S 45 A/S
M2 A/S 46 A/S
M3 A/S 47 A/S
M4 A/S 48 A/S
M5 A/S 49 A/S
M6 A/S 50 A/S
M7 A/S 51 A/S
M8 A/S 52 A/S
M9 A/S 53 A/S

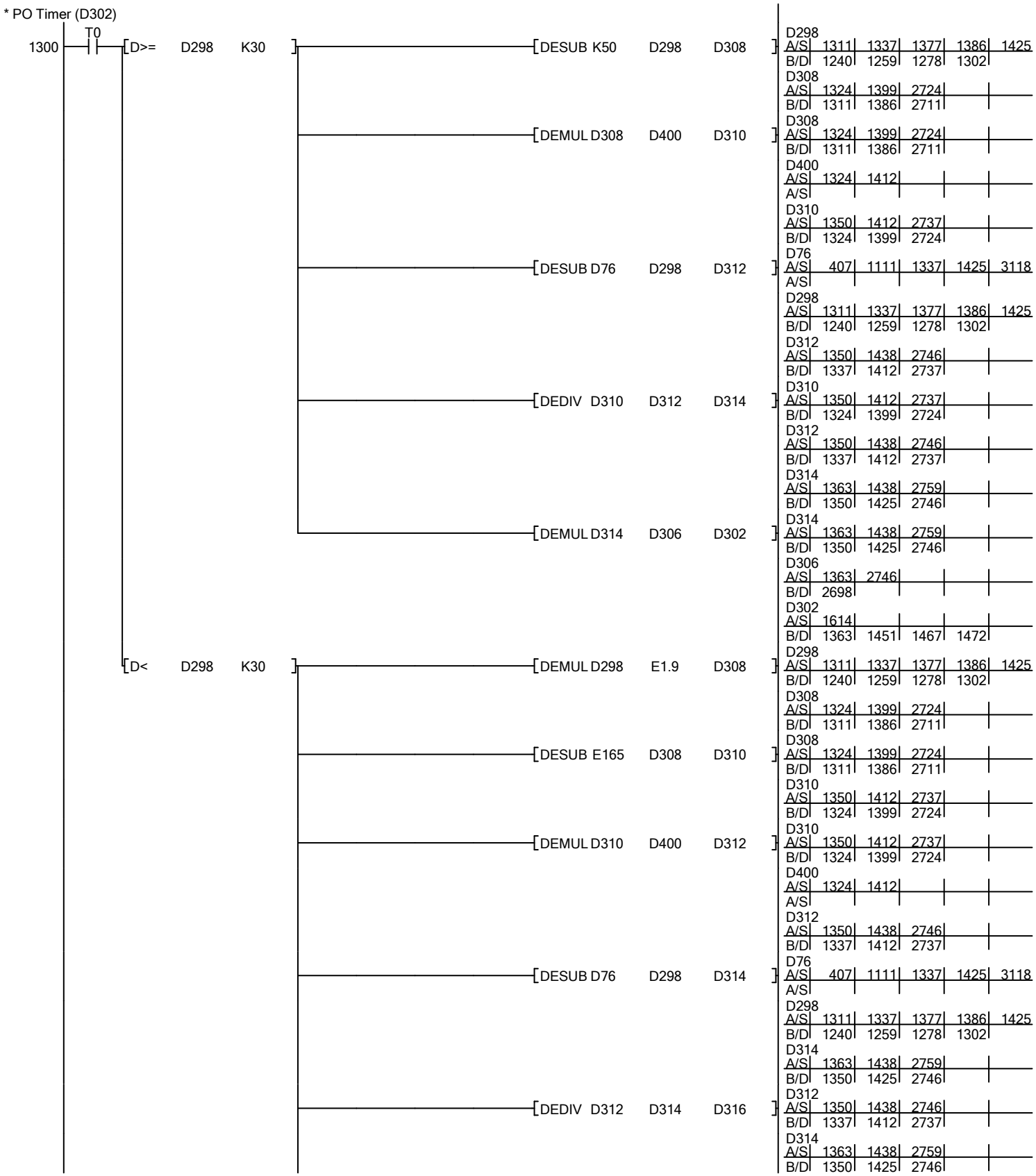
* Water Temperature Target Logic

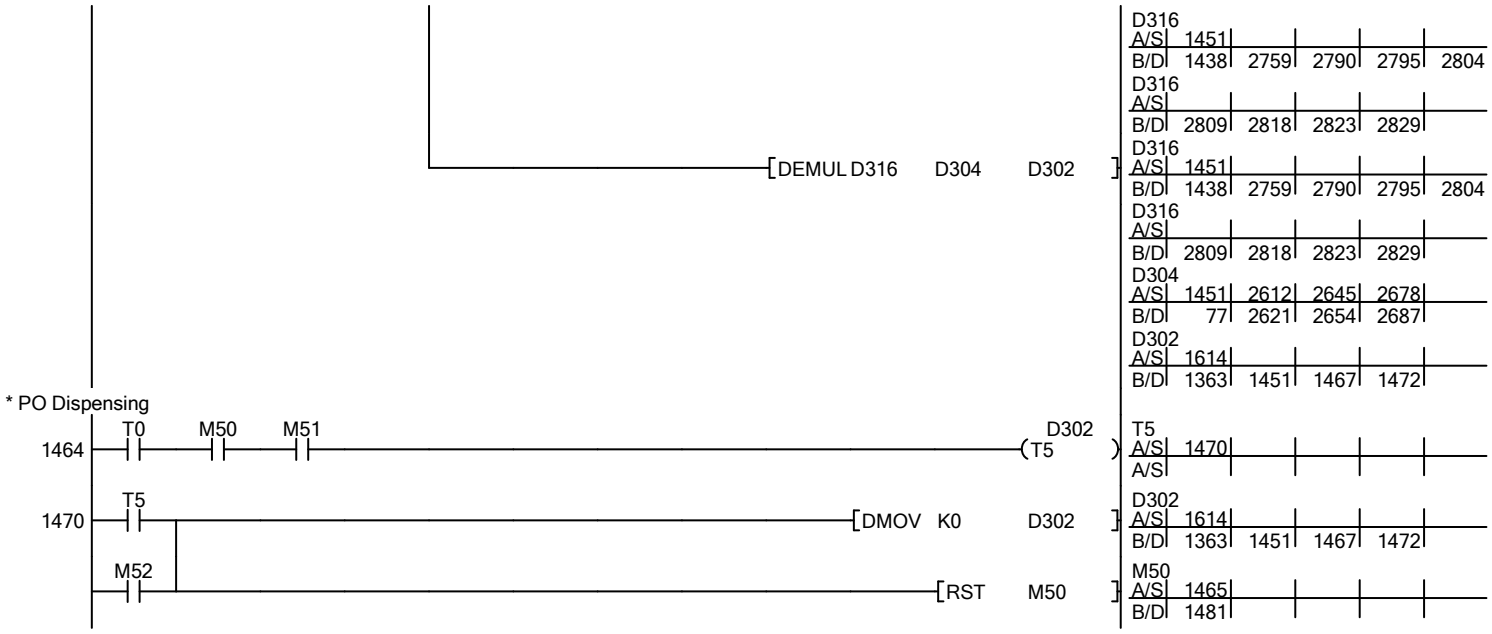


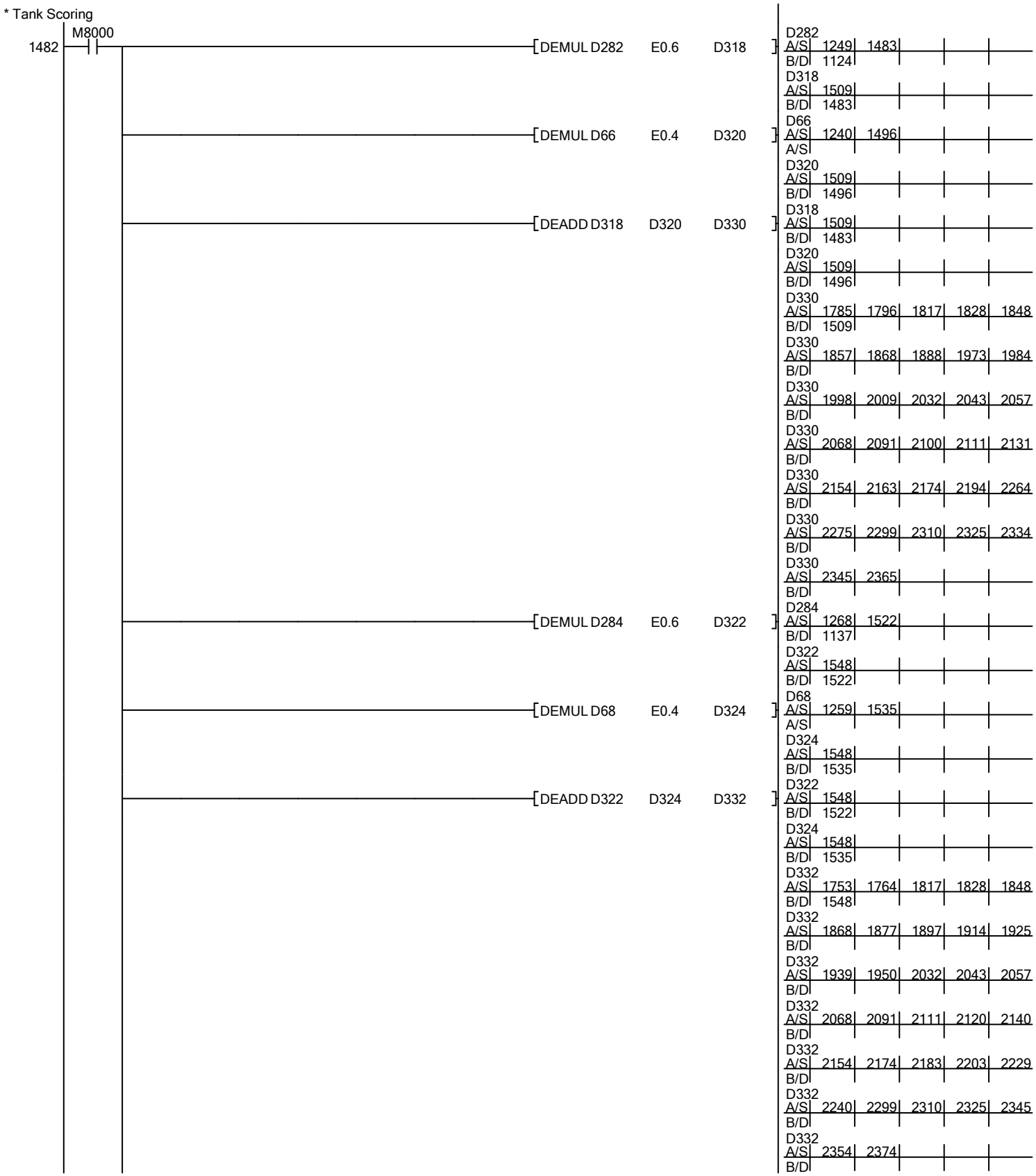

* Heater latching
949
M11
D> D220 K1
D<= D122 K55
MOV K3 D278
D>= D122 K55
MOV K2 D278
D>= D122 K80
MOV K1 D278
D>= D122 K95
MOV K0 D278
D< D220 K1
D> D122 K90
MOV K0 D278
D= D220 K1
D<= D122 K85
MOV K1 D278
D<= D122 K75
MOV K2 D278
D<= D122 K50
MOV K3 D278
D278
A/S 815
B/D 970 985 1000 1015 1050
A/S
B/D 1065 1080 1095
* Manual Heater Control
1100
M12
X006
Y010
X007
Y011
X010
Y012

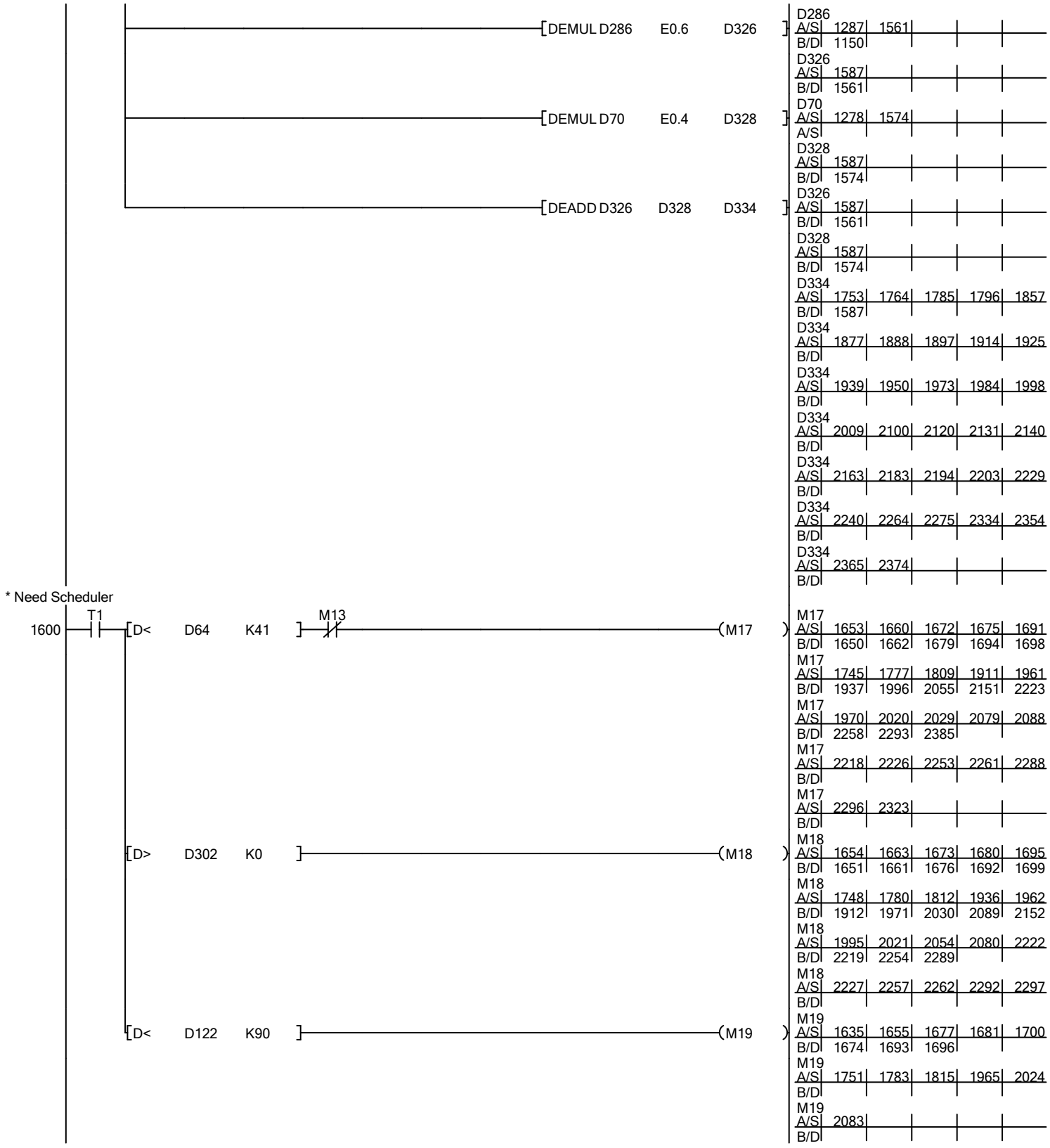

* Height to Mass
1110 T0
DEMUL D76 K1 D280
DEMUL D78 K1 D282
DEMUL D80 K1 D284
DEMUL D82 K1 D286
D76 A/S 407 1111 1337 1425 3118
D280 A/S 1164 1190 B/D 1111
D78 A/S 420 1124
D282 A/S 1249 1483 B/D 1124
D80 A/S 433 1137
D284 A/S 1268 1522 B/D 1137
D82 A/S 446 1150
D286 A/S 1287 1561 B/D 1150
* Dynamic Upper Thermal Boundary
1163 T0
DEMUL D280 D64 D288
DEMUL D74 E239.53 D290
DEADD D280 E239.53 D292
DEADD D288 D290 D294
DEDIV D294 D292 D296
D64 A/S 565 587 1164 1602 3108 B/D 369
D74 A/S 1177
D288 A/S 1203 B/D 1164
D290 A/S 1203 B/D 1177
D292 A/S 1216 B/D 1190
D294 A/S 1216 B/D 1203
D296 A/S 1229 3103 B/D 1216
1229 D> D296 K55
M13
M13 B/D 1611

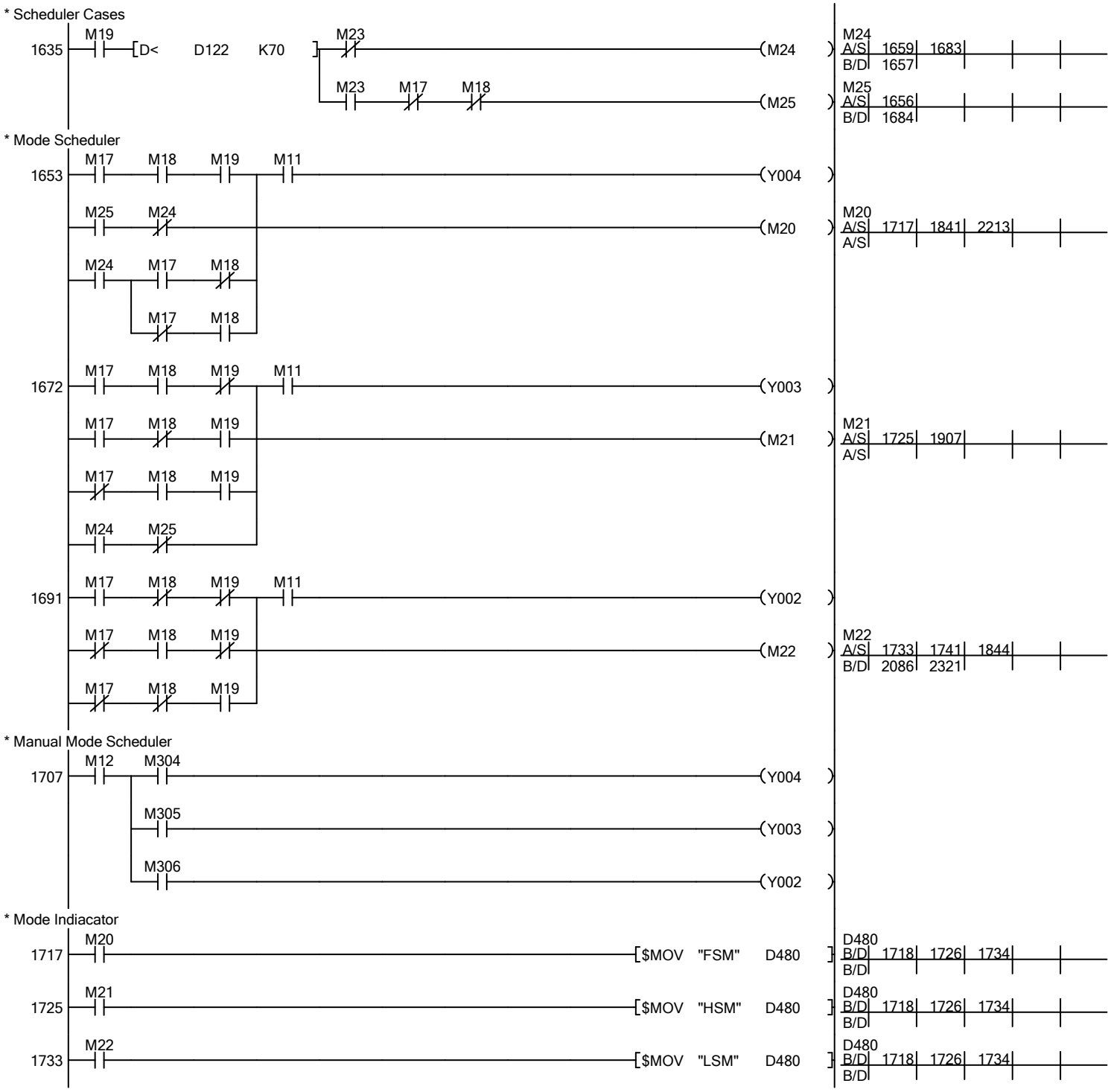

* PO Designation
1239
M14
DEMOVD66 D298
DEMOVD282 D300
1258
M15
DEMOVD68 D298
DEMOVD284 D300
1277
M16
DEMOVD70 D298
DEMOVD286 D300
1296
M14 M15 M16
M23
D66 A/S 1240 1496 A/S
D298 A/S 1311 1337 1377 1386 1425 B/D 1240 1259 1278 1302
D282 A/S 1249 1483 B/D 1124
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
D68 A/S 1259 1535 A/S
D298 A/S 1311 1337 1377 1386 1425 B/D 1240 1259 1278 1302
D284 A/S 1268 1522 B/D 1137
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
D70 A/S 1278 1574 A/S
D298 A/S 1311 1337 1377 1386 1425 B/D 1240 1259 1278 1302
D286 A/S 1287 1561 B/D 1150
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
M23 A/S 1649 1839 B/D 1646

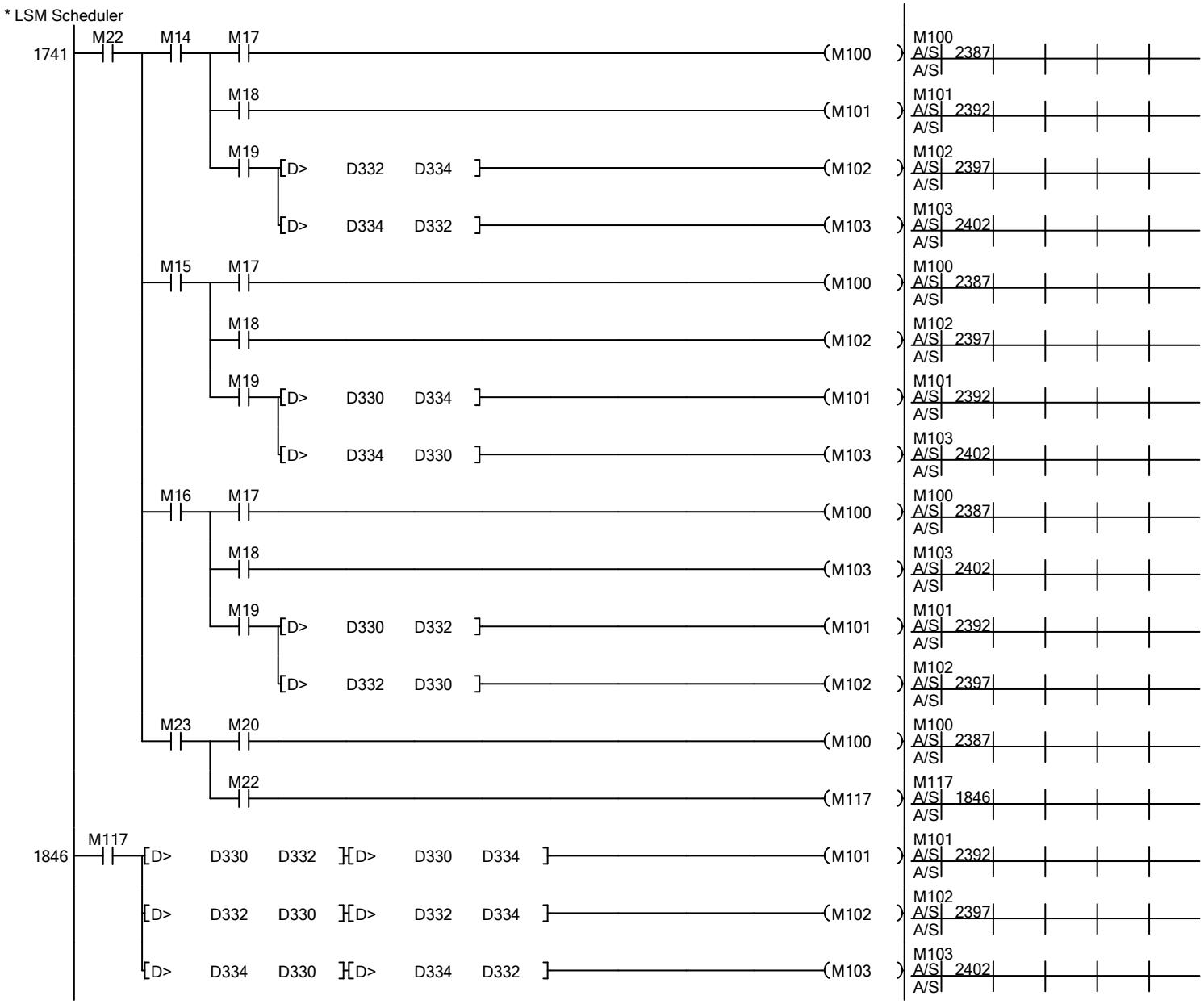

* PO Timer (D302)
1300
T0
D>= D298 K30
D< D298 K30
DESUB K50 D298 D308
DEMUL D308 D400 D310
DESUB D76 D298 D312
DEDIV D310 D312 D314
DEMUL D314 D306 D302
DEMUL D298 E1.9 D308
DESUB E165 D308 D310
DEMUL D310 D400 D312
DESUB D76 D298 D314
DEDIV D312 D314 D316

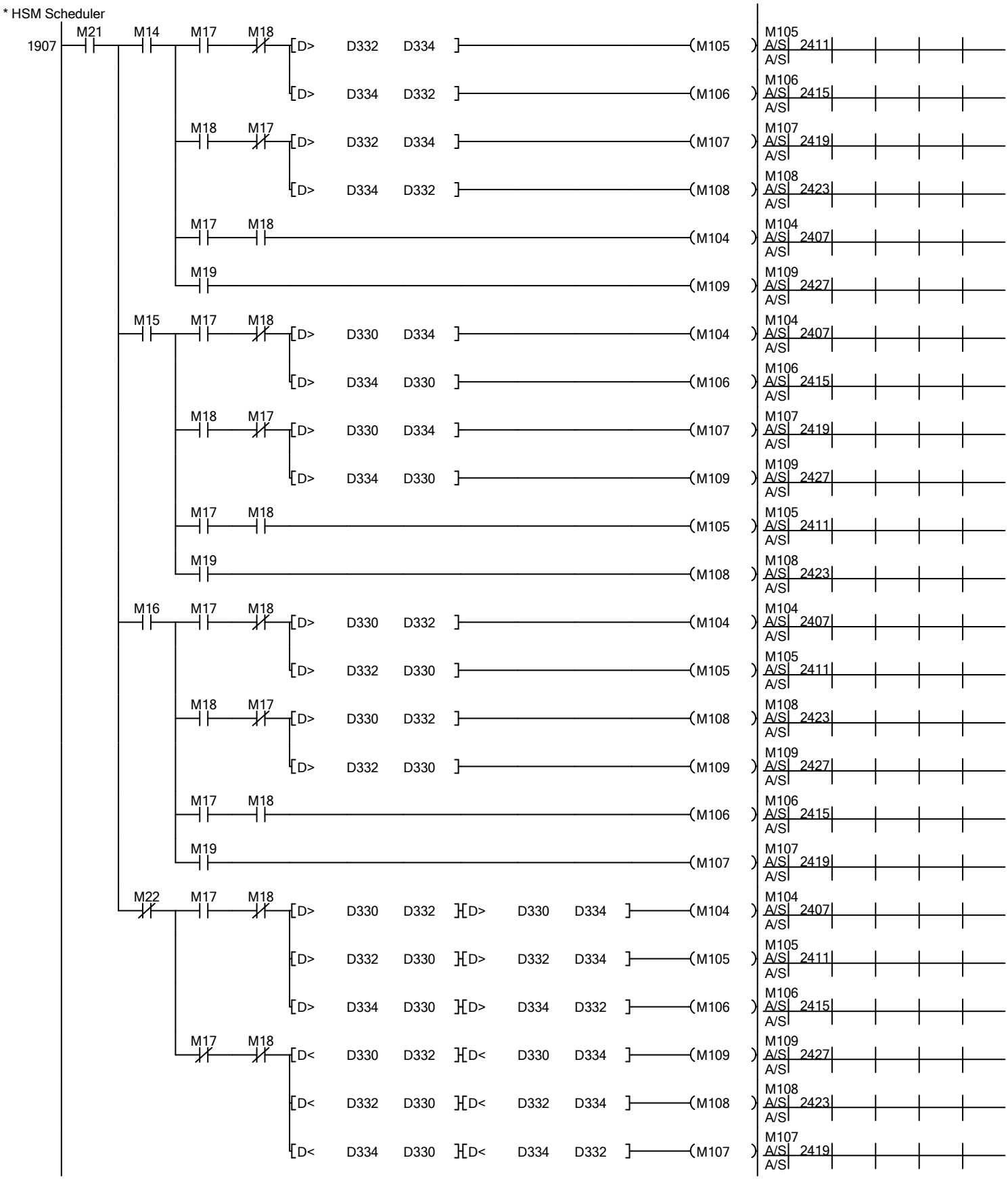

[DEMUL D316 D304 D302]
D316
A/S 1451
B/D 1438 2759 2790 2795 2804
D316
A/S
B/D 2809 2818 2823 2829
D316
A/S 1451
B/D 1438 2759 2790 2795 2804
D316
A/S
B/D 2809 2818 2823 2829
D304
A/S 1451 2612 2645 2678
B/D 77 2621 2654 2687
D302
A/S 1614
B/D 1363 1451 1467 1472
* PO Dispensing
1464 T0 M50 M51 (T5 D302)
T5
A/S 1470
A/S
1470 T5 [DMOV K0 D302]
D302
A/S 1614
B/D 1363 1451 1467 1472
M52 [RST M50]
M50
A/S 1465
B/D 1481

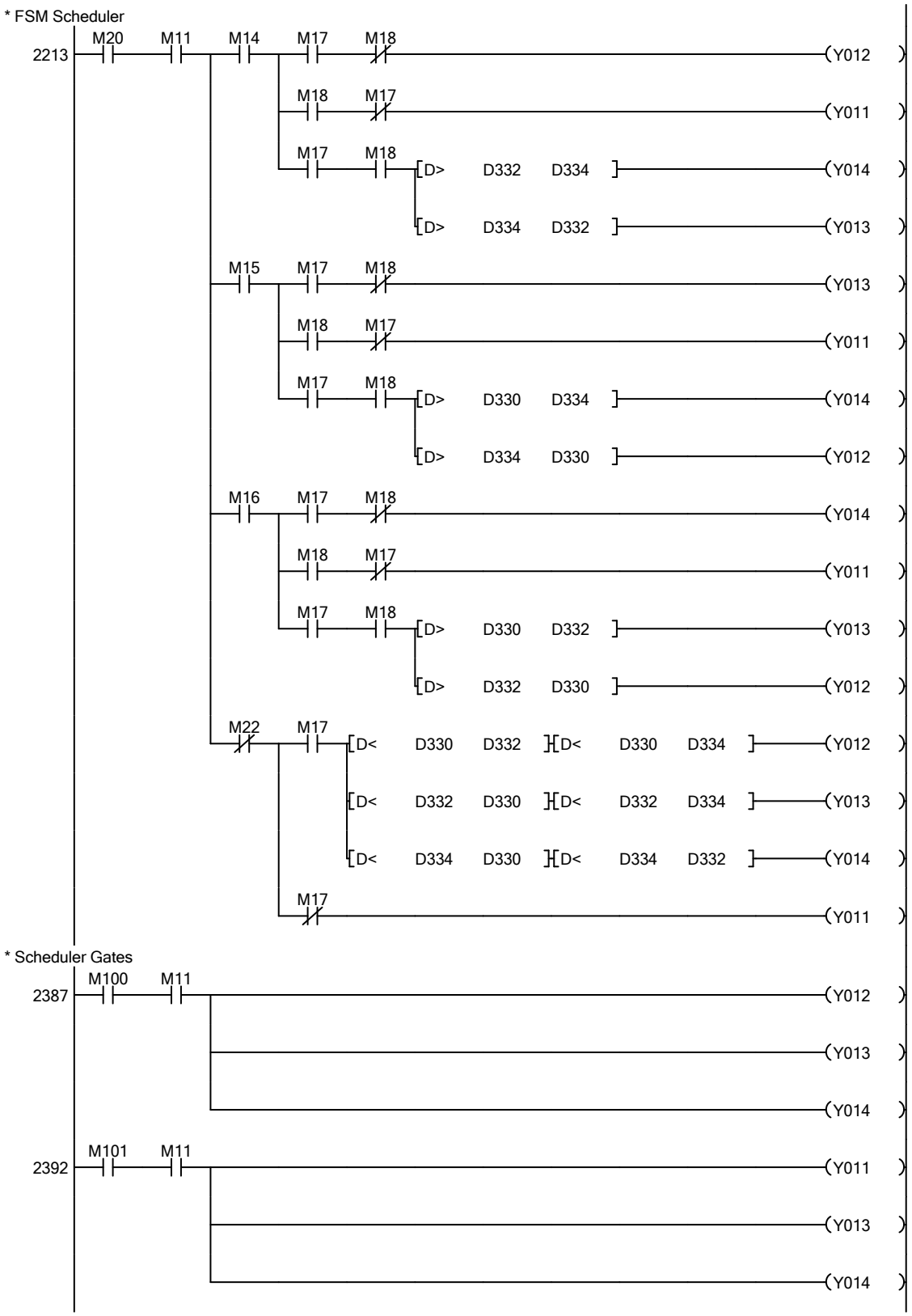

* Tank Scoring
M8000
1482
DEMUL D282 E0.6 D318
D282 A/S 1249 1483 B/D 1124
D318 A/S 1509 B/D 1483
DEMUL D66 E0.4 D320
D66 A/S 1240 1496 A/S
D320 A/S 1509 B/D 1496
DEADD D318 D320 D330
D318 A/S 1509 B/D 1483
D320 A/S 1509 B/D 1496
D330 A/S 1785 1796 1817 1828 1848 B/D 1509
D330 A/S 1857 1868 1888 1973 1984 B/D
D330 A/S 1998 2009 2032 2043 2057 B/D
D330 A/S 2068 2091 2100 2111 2131 B/D
D330 A/S 2154 2163 2174 2194 2264 B/D
D330 A/S 2275 2299 2310 2325 2334 B/D
D330 A/S 2345 2365 B/D
DEMUL D284 E0.6 D322
D284 A/S 1268 1522 B/D 1137
D322 A/S 1548 B/D 1522
DEMUL D68 E0.4 D324
D68 A/S 1259 1535 A/S
D324 A/S 1548 B/D 1535
DEADD D322 D324 D332
D322 A/S 1548 B/D 1522
D324 A/S 1548 B/D 1535
D332 A/S 1753 1764 1817 1828 1848 B/D 1548
D332 A/S 1868 1877 1897 1914 1925 B/D
D332 A/S 1939 1950 2032 2043 2057 B/D
D332 A/S 2068 2091 2111 2120 2140 B/D
D332 A/S 2154 2174 2183 2203 2229 B/D
D332 A/S 2240 2299 2310 2325 2345 B/D
D332 A/S 2354 2374 B/D

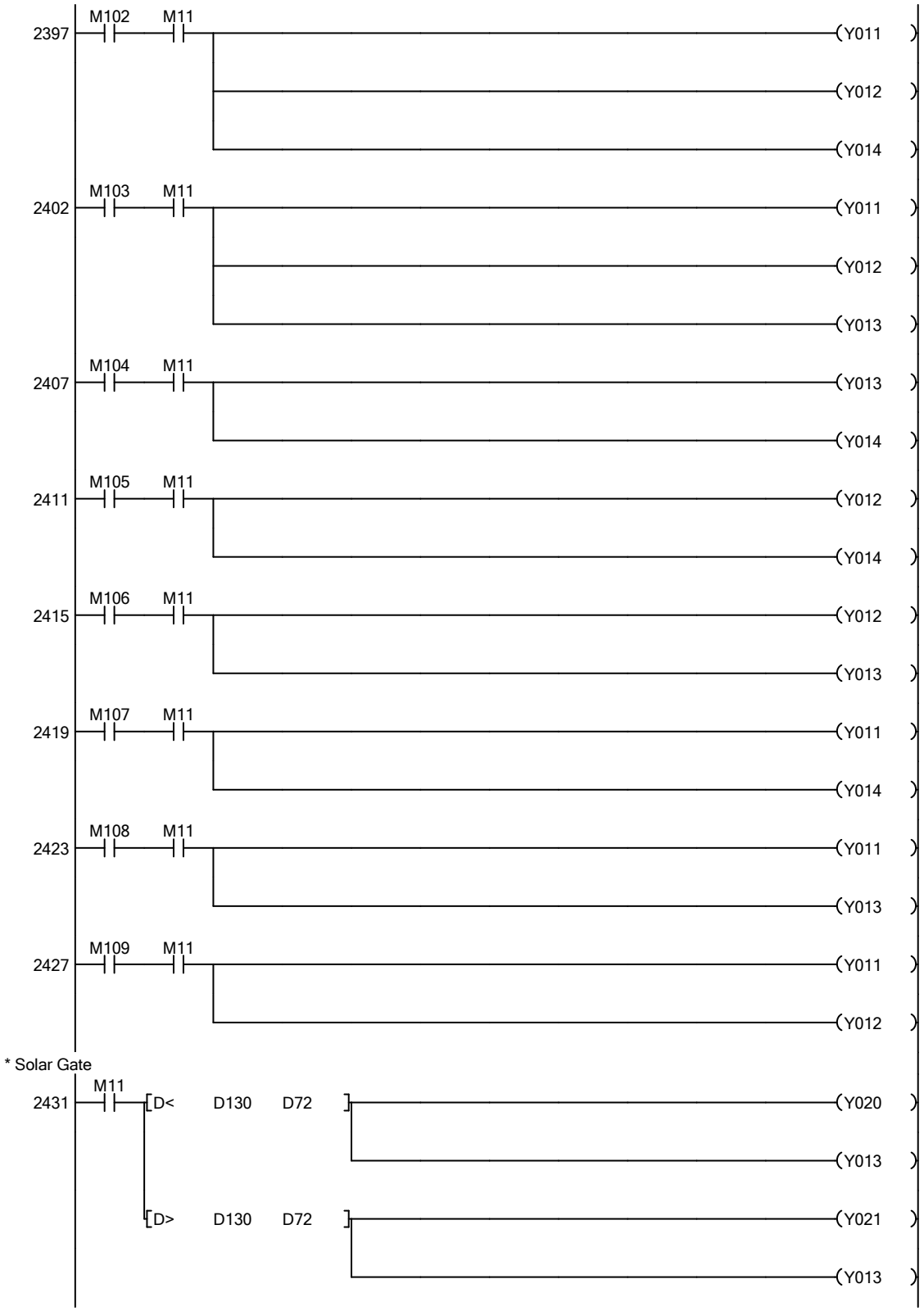

[DEMUL D286 E0.6 D326]
[DEMUL D70 E0.4 D328]
[DEADD D326 D328 D334]
* Need Scheduler
1600
T1
[D< D64 K41]
M13
(M17)
[D> D302 K0]
(M18)
[D< D122 K90]
(M19)
D286 A/S 1287 1561 B/D 1150
D326 A/S 1587 B/D 1561
D70 A/S 1278 1574 A/S
D328 A/S 1587 B/D 1574
D326 A/S 1587 B/D 1561
D328 A/S 1587 B/D 1574
D334 A/S 1753 1764 1785 1796 1857 B/D 1587
D334 A/S 1877 1888 1897 1914 1925 B/D
D334 A/S 1939 1950 1973 1984 1998 B/D
D334 A/S 2009 2100 2120 2131 2140 B/D
D334 A/S 2163 2183 2194 2203 2229 B/D
D334 A/S 2240 2264 2275 2334 2354 B/D
D334 A/S 2365 2374 B/D
M17 A/S 1653 1660 1672 1675 1691 B/D 1650 1662 1679 1694 1698
M17 A/S 1745 1777 1809 1911 1961 B/D 1937 1996 2055 2151 2223
M17 A/S 1970 2020 2029 2079 2088 B/D 2258 2293 2385
M17 A/S 2218 2226 2253 2261 2288 B/D
M17 A/S 2296 2323 B/D
M18 A/S 1654 1663 1673 1680 1695 B/D 1651 1661 1676 1692 1699
M18 A/S 1748 1780 1812 1936 1962 B/D 1912 1971 2030 2089 2152
M18 A/S 1995 2021 2054 2080 2222 B/D 2219 2254 2289
M18 A/S 2227 2257 2262 2292 2297 B/D
M19 A/S 1635 1655 1677 1681 1700 B/D 1674 1693 1696
M19 A/S 1751 1783 1815 1965 2024 B/D
M19 A/S 2083 B/D

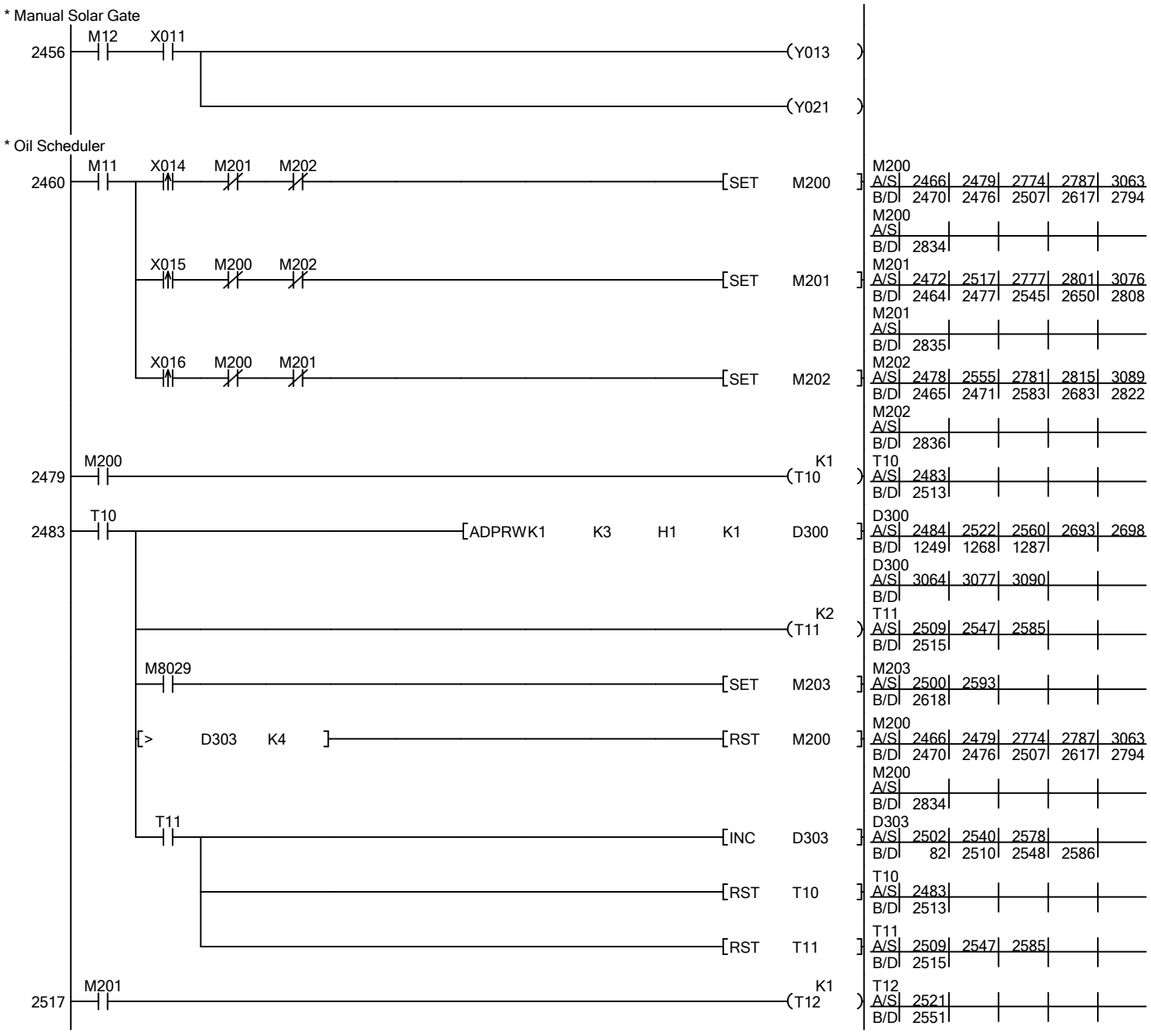

* Scheduler Cases
1635
M19
D< D122 K70
M23
M24
M23 M17 M18
M25
M24 A/S 1659 1683
B/D 1657
M25 A/S 1656
B/D 1684
* Mode Scheduler
1653
M17 M18 M19 M11
Y004
M25 M24
M20
M20 A/S 1717 1841 2213
A/S
M24 M17 M18
M17 M18
1672
M17 M18 M19 M11
Y003
M17 M18 M19
M21
M21 A/S 1725 1907
A/S
M17 M18 M19
M24 M25
1691
M17 M18 M19 M11
Y002
M17 M18 M19
M22
M22 A/S 1733 1741 1844
B/D 2086 2321
M17 M18 M19
* Manual Mode Scheduler
1707
M12 M304
Y004
M305
Y003
M306
Y002
* Mode Indiacator
1717
M20
$MOV "FSM" D480
1725
M21
$MOV "HSM" D480
1733
M22
$MOV "LSM" D480
D480 B/D 1718 1726 1734
B/D

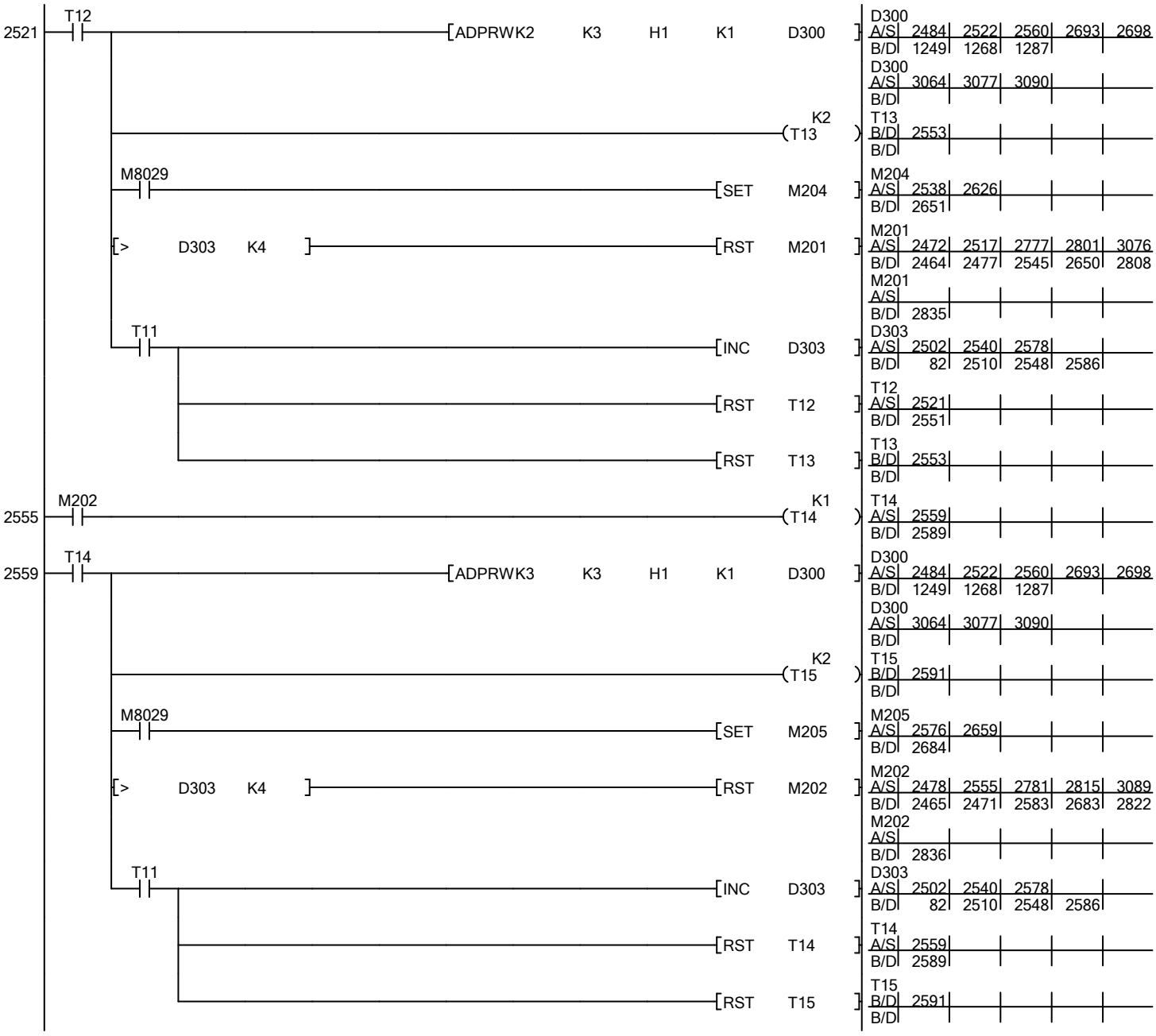

* LSM Scheduler
1741
M22
M14
M17
M100
M100 A/S 2387 A/S
M18
M101
M101 A/S 2392 A/S
M19
D> D332 D334
M102
M102 A/S 2397 A/S
D> D334 D332
M103
M103 A/S 2402 A/S
M15
M17
M100
M100 A/S 2387 A/S
M18
M102
M102 A/S 2397 A/S
M19
D> D330 D334
M101
M101 A/S 2392 A/S
D> D334 D330
M103
M103 A/S 2402 A/S
M16
M17
M100
M100 A/S 2387 A/S
M18
M103
M103 A/S 2402 A/S
M19
D> D330 D332
M101
M101 A/S 2392 A/S
D> D332 D330
M102
M102 A/S 2397 A/S
M23
M20
M100
M100 A/S 2387 A/S
M22
M117
M117 A/S 1846 A/S
1846
M117
D> D330 D332
D> D330 D334
M101
M101 A/S 2392 A/S
D> D332 D330
D> D332 D334
M102
M102 A/S 2397 A/S
D> D334 D330
D> D334 D332
M103
M103 A/S 2402 A/S

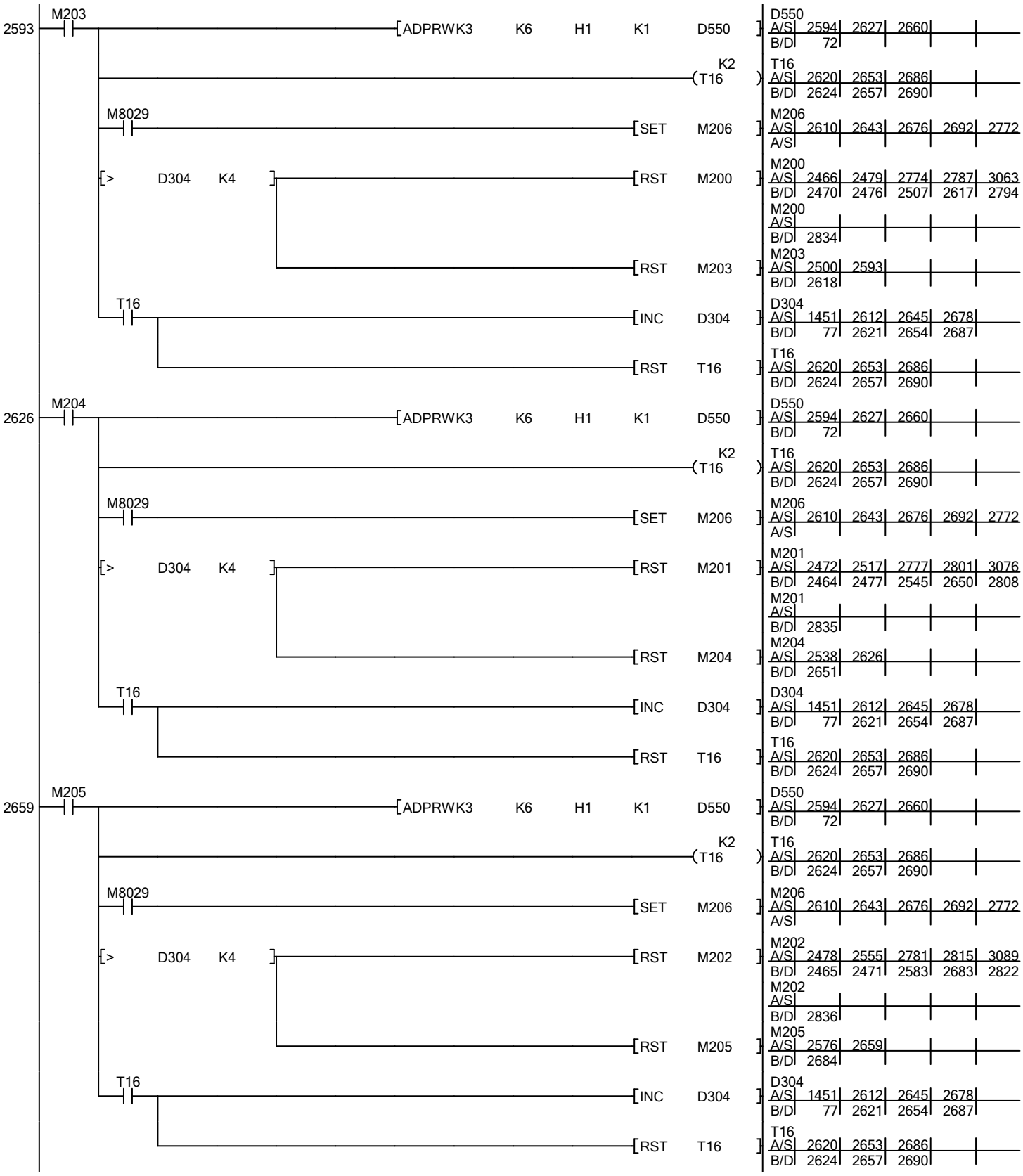

* HSM Scheduler
1907
M21 M14 M17 M18 [D> D332 D334] (M105) M105 A/S 2411 A/S
[D> D334 D332] (M106) M106 A/S 2415 A/S
M18 M17 [D> D332 D334] (M107) M107 A/S 2419 A/S
[D> D334 D332] (M108) M108 A/S 2423 A/S
M17 M18 (M104) M104 A/S 2407 A/S
M19 (M109) M109 A/S 2427 A/S
M15 M17 M18 [D> D330 D334] (M104) M104 A/S 2407 A/S
[D> D334 D330] (M106) M106 A/S 2415 A/S
M18 M17 [D> D330 D334] (M107) M107 A/S 2419 A/S
[D> D334 D330] (M109) M109 A/S 2427 A/S
M17 M18 (M105) M105 A/S 2411 A/S
M19 (M108) M108 A/S 2423 A/S
M16 M17 M18 [D> D330 D332] (M104) M104 A/S 2407 A/S
[D> D332 D330] (M105) M105 A/S 2411 A/S
M18 M17 [D> D330 D332] (M108) M108 A/S 2423 A/S
[D> D332 D330] (M109) M109 A/S 2427 A/S
M17 M18 (M106) M106 A/S 2415 A/S
M19 (M107) M107 A/S 2419 A/S
M22 M17 M18 [D> D330 D332][D> D330 D334] (M104) M104 A/S 2407 A/S
[D> D332 D330][D> D332 D334] (M105) M105 A/S 2411 A/S
[D> D334 D330][D> D334 D332] (M106) M106 A/S 2415 A/S
M17 M18 [D< D330 D332][D< D330 D334] (M109) M109 A/S 2427 A/S
[D< D332 D330][D< D332 D334] (M108) M108 A/S 2423 A/S
[D< D334 D330][D< D334 D332] (M107) M107 A/S 2419 A/S

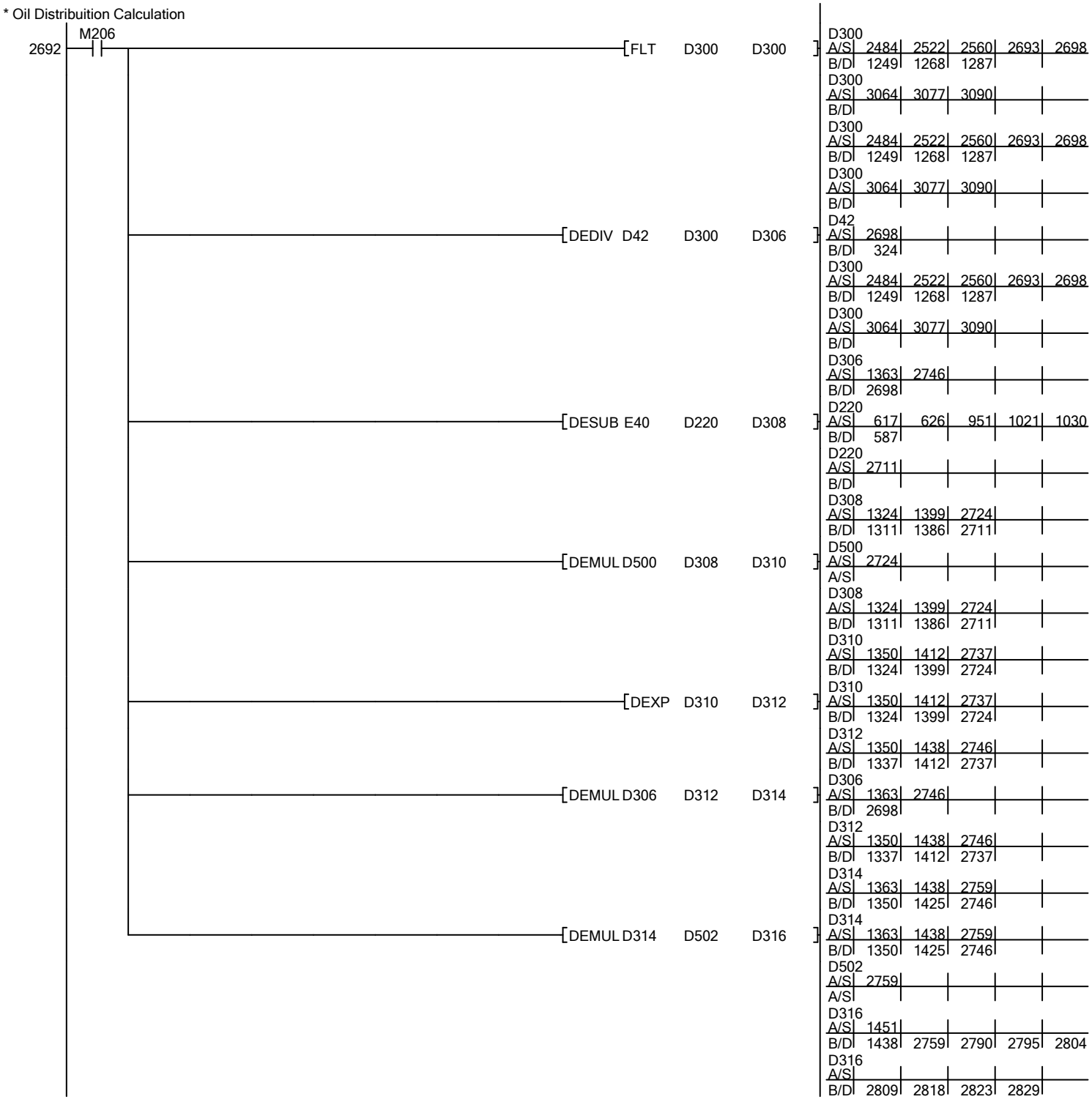

* FSM Scheduler
2213
M20 M11 M14 M17 M18 (Y012)
M18 M17 (Y011)
M17 M18 [D> D332 D334] (Y014)
[D> D334 D332] (Y013)
M15 M17 M18 (Y013)
M18 M17 (Y011)
M17 M18 [D> D330 D334] (Y014)
[D> D334 D330] (Y012)
M16 M17 M18 (Y014)
M18 M17 (Y011)
M17 M18 [D> D330 D332] (Y013)
[D> D332 D330] (Y012)
M22 M17 [D< D330 D332][D< D330 D334] (Y012)
[D< D332 D330][D< D332 D334] (Y013)
[D< D334 D330][D< D334 D332] (Y014)
M17 (Y011)
* Scheduler Gates
2387
M100 M11 (Y012)
(Y013)
(Y014)
2392
M101 M11 (Y011)
(Y013)
(Y014)

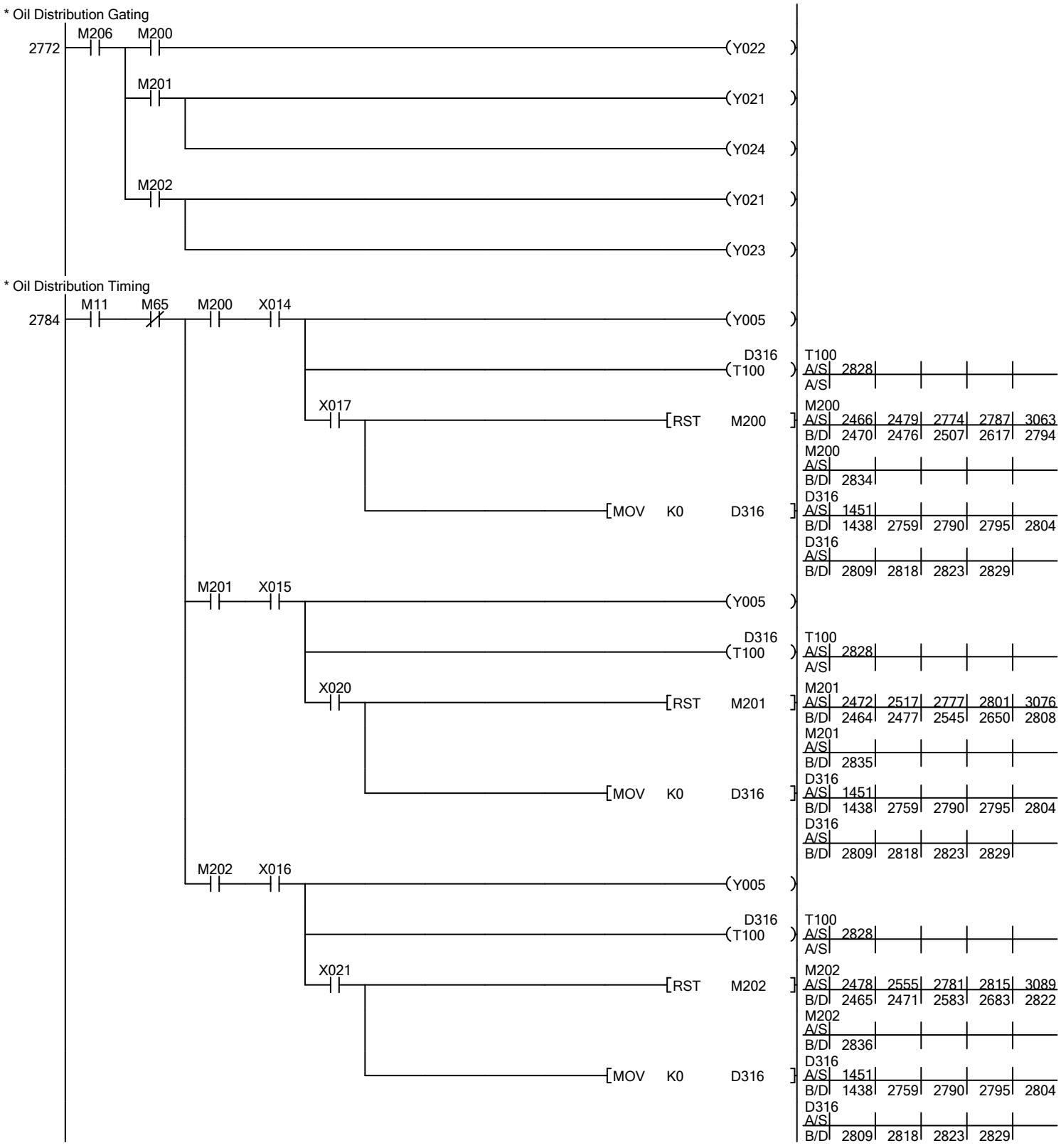

2397 M102 M11 Y011 Y012 Y014
2402 M103 M11 Y011 Y012 Y013
2407 M104 M11 Y013 Y014
2411 M105 M11 Y012 Y014
2415 M106 M11 Y012 Y013
2419 M107 M11 Y011 Y014
2423 M108 M11 Y011 Y013
2427 M109 M11 Y011 Y012
* Solar Gate
2431 M11 D< D130 D72 Y020 Y013
D> D130 D72 Y021 Y013

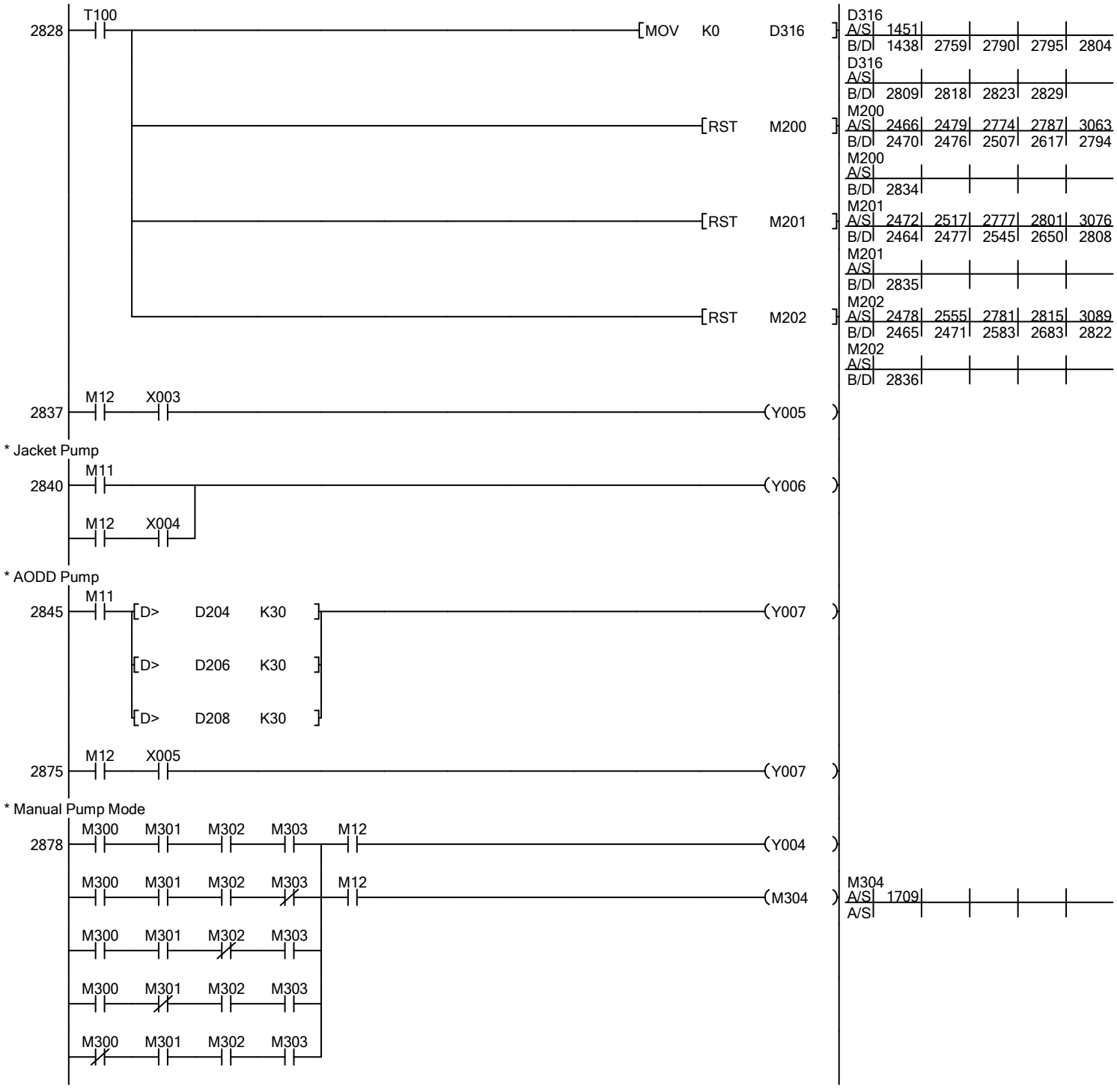

* Manual Solar Gate
2456
M12
X011
Y013
Y021
* Oil Scheduler
2460
M11
X014
M201
M202
SET M200
X015
M200
M202
SET M201
X016
M200
M201
SET M202
2479
M200
K1
T10
2483
T10
ADPRWK1 K3 H1 K1 D300
K2
T11
M8029
SET M203
> D303 K4
RST M200
T11
INC D303
RST T10
RST T11
2517
M201
K1
T12
M200 A/S 2466 2479 2774 2787 3063 B/D 2470 2476 2507 2617 2794
M200 A/S B/D 2834
M201 A/S 2472 2517 2777 2801 3076 B/D 2464 2477 2545 2650 2808
M201 A/S B/D 2835
M202 A/S 2478 2555 2781 2815 3089 B/D 2465 2471 2583 2683 2822
M202 A/S B/D 2836
T10 A/S 2483 B/D 2513
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
T11 A/S 2509 2547 2585 B/D 2515
M203 A/S 2500 2593 B/D 2618
M200 A/S 2466 2479 2774 2787 3063 B/D 2470 2476 2507 2617 2794
M200 A/S B/D 2834
D303 A/S 2502 2540 2578 B/D 82 2510 2548 2586
T10 A/S 2483 B/D 2513
T11 A/S 2509 2547 2585 B/D 2515
T12 A/S 2521 B/D 2551

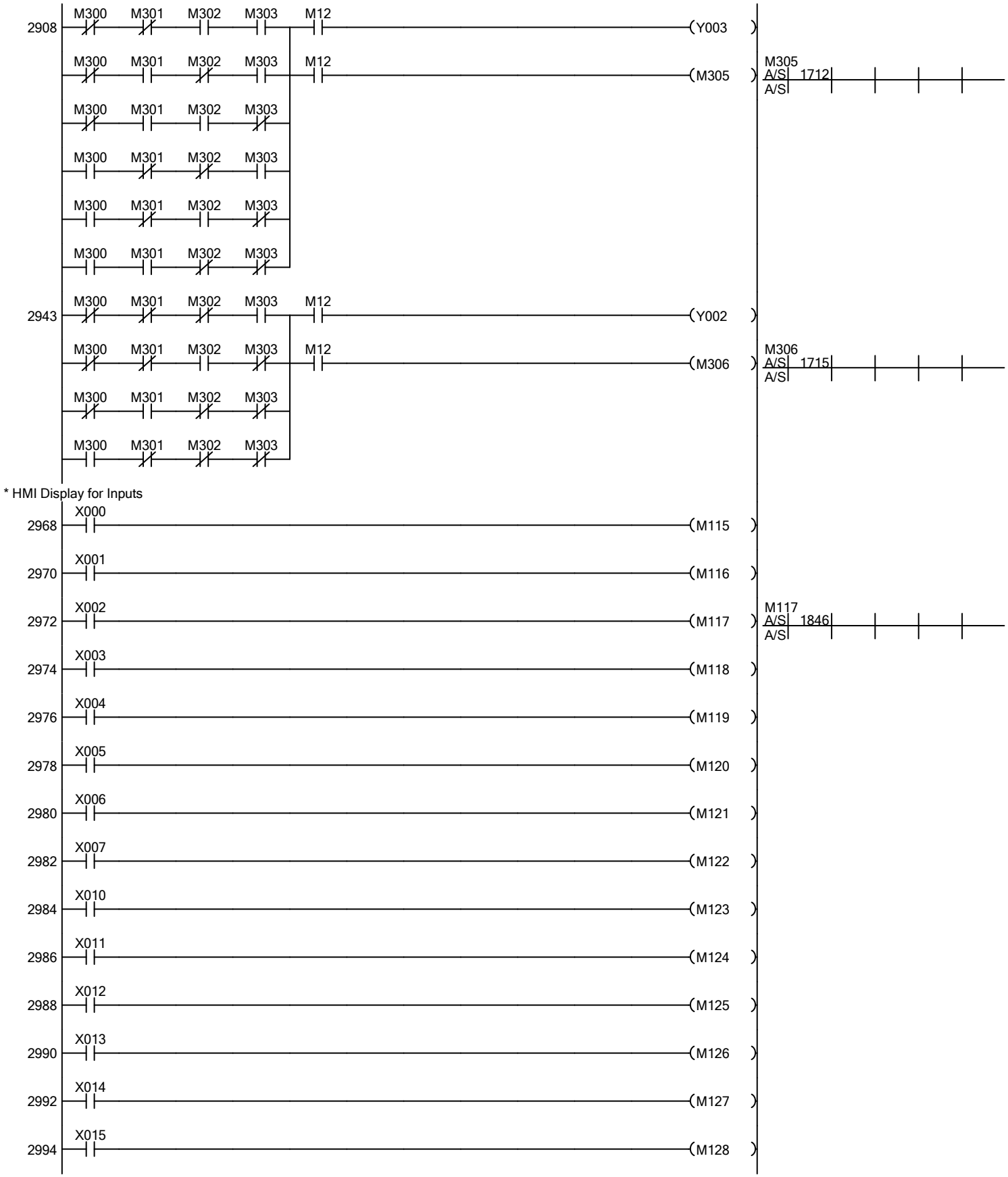

2521
T12
ADPRW K2 K3 H1 K1 D300
K2
T13
M8029
SET M204
> D303 K4
RST M201
T11
INC D303
RST T12
RST T13
2555
M202
K1
T14
2559
T14
ADPRW K3 K3 H1 K1 D300
K2
T15
M8029
SET M205
> D303 K4
RST M202
T11
INC D303
RST T14
RST T15
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
T13 B/D 2553 B/D
M204 A/S 2538 2626 B/D 2651
M201 A/S 2472 2517 2777 2801 3076 B/D 2464 2477 2545 2650 2808
M201 A/S B/D 2835
D303 A/S 2502 2540 2578 B/D 82 2510 2548 2586
T12 A/S 2521 B/D 2551
T13 B/D 2553 B/D
T14 A/S 2559 B/D 2589
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
T15 B/D 2591 B/D
M205 A/S 2576 2659 B/D 2684
M202 A/S 2478 2555 2781 2815 3089 B/D 2465 2471 2583 2683 2822
M202 A/S B/D 2836
D303 A/S 2502 2540 2578 B/D 82 2510 2548 2586
T14 A/S 2559 B/D 2589
T15 B/D 2591 B/D

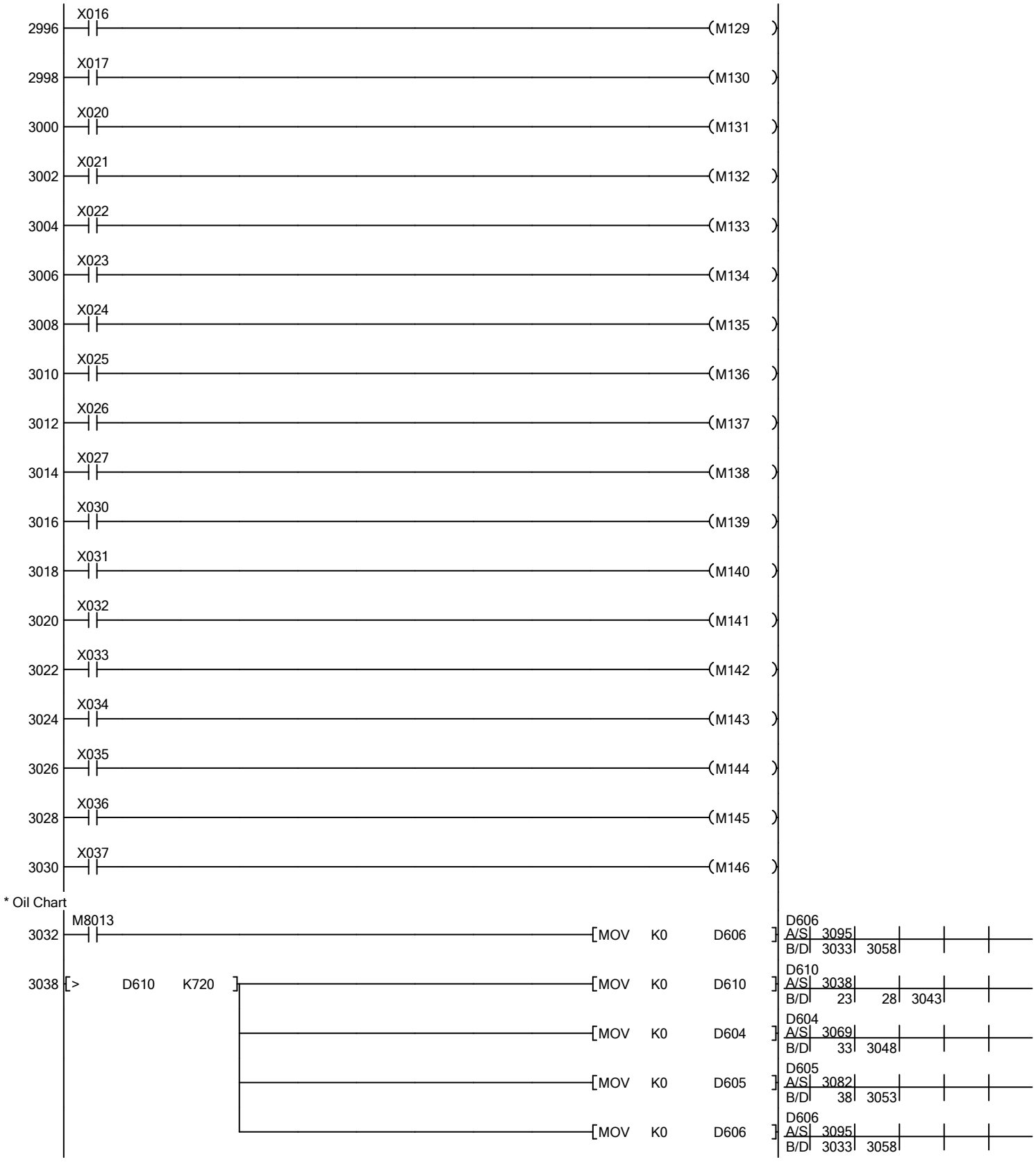

2593
M203
ADPRWK3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M200
RST M203
T16
INC D304
RST T16
2626
M204
ADPRWK3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M201
RST M204
T16
INC D304
RST T16
2659
M205
ADPRWK3 K6 H1 K1 D550
K2
T16
M8029
SET M206
> D304 K4
RST M202
RST M205
T16
INC D304
RST T16
D550 A/S 2594 2627 2660 B/D 72
T16 A/S 2620 2653 2686 B/D 2624 2657 2690
M206 A/S 2610 2643 2676 2692 2772 A/S
M200 A/S 2466 2479 2774 2787 3063 B/D 2470 2476 2507 2617 2794
M200 A/S B/D 2834
M203 A/S 2500 2593 B/D 2618
D304 A/S 1451 2612 2645 2678 B/D 77 2621 2654 2687
T16 A/S 2620 2653 2686 B/D 2624 2657 2690
D550 A/S 2594 2627 2660 B/D 72
T16 A/S 2620 2653 2686 B/D 2624 2657 2690
M206 A/S 2610 2643 2676 2692 2772 A/S
M201 A/S 2472 2517 2777 2801 3076 B/D 2464 2477 2545 2650 2808
M201 A/S B/D 2835
M204 A/S 2538 2626 B/D 2651
D304 A/S 1451 2612 2645 2678 B/D 77 2621 2654 2687
T16 A/S 2620 2653 2686 B/D 2624 2657 2690
D550 A/S 2594 2627 2660 B/D 72
T16 A/S 2620 2653 2686 B/D 2624 2657 2690
M206 A/S 2610 2643 2676 2692 2772 A/S
M202 A/S 2478 2555 2781 2815 3089 B/D 2465 2471 2583 2683 2822
M202 A/S B/D 2836
M205 A/S 2576 2659 B/D 2684
D304 A/S 1451 2612 2645 2678 B/D 77 2621 2654 2687
T16 A/S 2620 2653 2686 B/D 2624 2657 2690

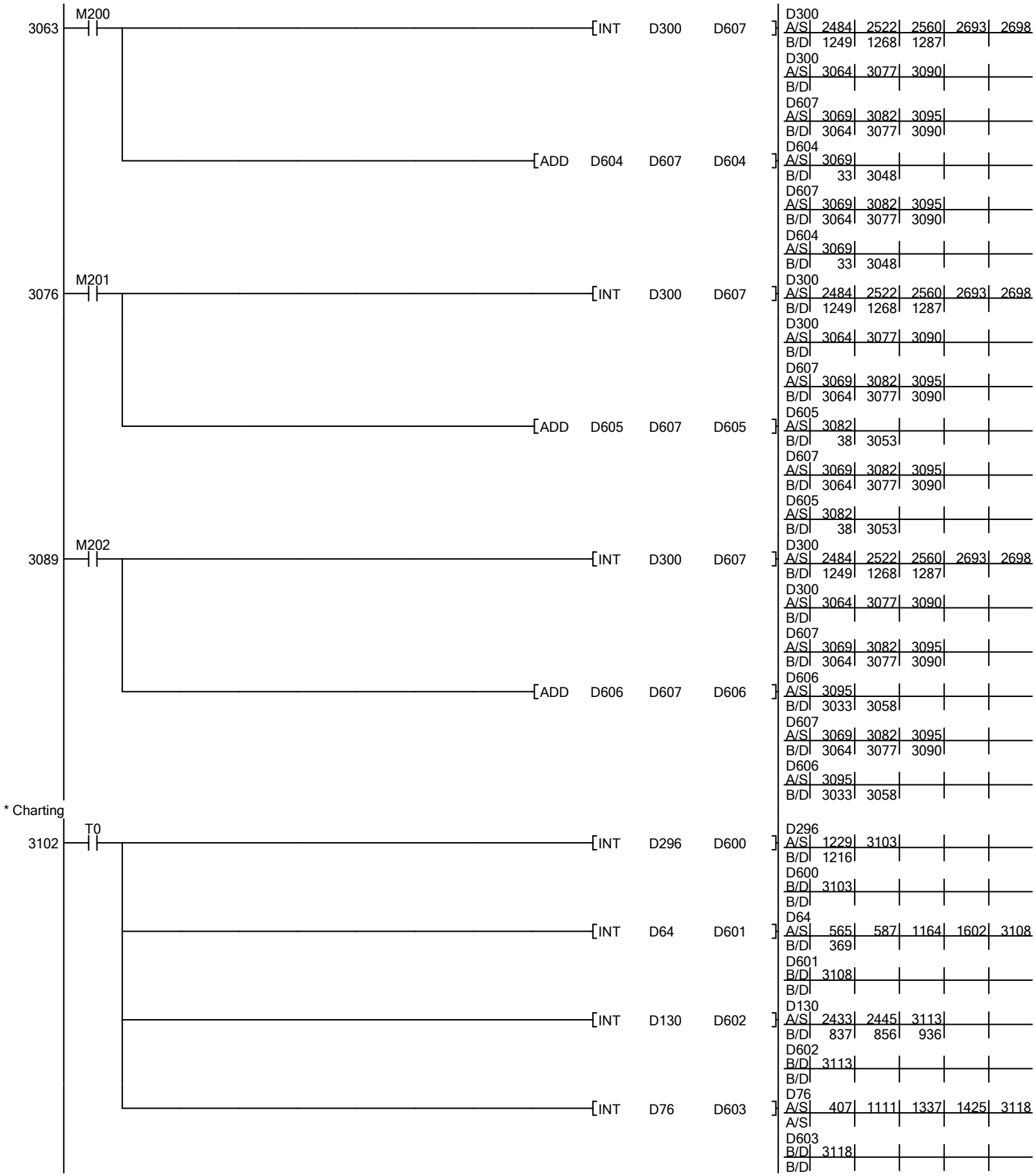

* Oil Distribuition Calculation
M206
2692
FLT D300 D300
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
DEDIV D42 D300 D306
D42 A/S 2698 B/D 324
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
D306 A/S 1363 2746 B/D 2698
DESUB E40 D220 D308
D220 A/S 617 626 951 1021 1030 B/D 587
D220 A/S 2711 B/D
D308 A/S 1324 1399 2724 B/D 1311 1386 2711
DEMUL D500 D308 D310
D500 A/S 2724 A/S
D308 A/S 1324 1399 2724 B/D 1311 1386 2711
D310 A/S 1350 1412 2737 B/D 1324 1399 2724
DEXP D310 D312
D310 A/S 1350 1412 2737 B/D 1324 1399 2724
D312 A/S 1350 1438 2746 B/D 1337 1412 2737
DEMUL D306 D312 D314
D306 A/S 1363 2746 B/D 2698
D312 A/S 1350 1438 2746 B/D 1337 1412 2737
D314 A/S 1363 1438 2759 B/D 1350 1425 2746
DEMUL D314 D502 D316
D314 A/S 1363 1438 2759 B/D 1350 1425 2746
D502 A/S 2759 A/S
D316 A/S 1451 B/D 1438 2759 2790 2795 2804
D316 A/S B/D 2809 2818 2823 2829

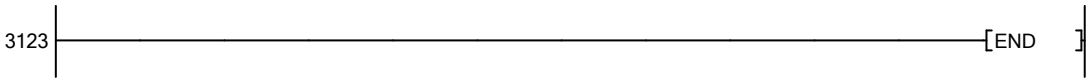

* Oil Distribution Gating
2772
M206
M200
Y022
M201
Y021
Y024
M202
Y021
Y023
* Oil Distribution Timing
2784
M11
M65
M200
X014
Y005
D316
T100
T100
A/S 2828
A/S
X017
RST M200
M200
A/S 2466 2479 2774 2787 3063
B/D 2470 2476 2507 2617 2794
M200
A/S
B/D 2834
MOV K0 D316
D316
A/S 1451
B/D 1438 2759 2790 2795 2804
D316
A/S
B/D 2809 2818 2823 2829
M201
X015
Y005
D316
T100
T100
A/S 2828
A/S
X020
RST M201
M201
A/S 2472 2517 2777 2801 3076
B/D 2464 2477 2545 2650 2808
M201
A/S
B/D 2835
MOV K0 D316
D316
A/S 1451
B/D 1438 2759 2790 2795 2804
D316
A/S
B/D 2809 2818 2823 2829
M202
X016
Y005
D316
T100
T100
A/S 2828
A/S
X021
RST M202
M202
A/S 2478 2555 2781 2815 3089
B/D 2465 2471 2583 2683 2822
M202
A/S
B/D 2836
MOV K0 D316
D316
A/S 1451
B/D 1438 2759 2790 2795 2804
D316
A/S
B/D 2809 2818 2823 2829


2828
T100
MOV K0 D316
RST M200
RST M201
RST M202
D316
A/S 1451
B/D 1438 2759 2790 2795 2804
D316
A/S
B/D 2809 2818 2823 2829
M200
A/S 2466 2479 2774 2787 3063
B/D 2470 2476 2507 2617 2794
M200
A/S
B/D 2834
M201
A/S 2472 2517 2777 2801 3076
B/D 2464 2477 2545 2650 2808
M201
A/S
B/D 2835
M202
A/S 2478 2555 2781 2815 3089
B/D 2465 2471 2583 2683 2822
M202
A/S
B/D 2836
2837
M12 X003
Y005
* Jacket Pump
2840
M11
M12 X004
Y006
* AODD Pump
2845
M11
D> D204 K30
D> D206 K30
D> D208 K30
Y007
2875
M12 X005
Y007
* Manual Pump Mode
2878
M300 M301 M302 M303 M12
Y004
M300 M301 M302 M303 M12
M304
M304
A/S 1709
A/S
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303


2908
M300 M301 M302 M303 M12
Y003
M300 M301 M302 M303 M12
M305
M305
A/S 1712
A/S
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
M300 M301 M302 M303
2943
M300 M301 M302 M303 M12
Y002
M300 M301 M302 M303 M12
M306
M306
A/S 1715
A/S
M300 M301 M302 M303
M300 M301 M302 M303
* HMI Display for Inputs
2968
X000
M115
2970
X001
M116
2972
X002
M117
M117
A/S 1846
A/S
2974
X003
M118
2976
X004
M119
2978
X005
M120
2980
X006
M121
2982
X007
M122
2984
X010
M123
2986
X011
M124
2988
X012
M125
2990
X013
M126
2992
X014
M127
2994
X015
M128


2996 X016 M129
2998 X017 M130
3000 X020 M131
3002 X021 M132
3004 X022 M133
3006 X023 M134
3008 X024 M135
3010 X025 M136
3012 X026 M137
3014 X027 M138
3016 X030 M139
3018 X031 M140
3020 X032 M141
3022 X033 M142
3024 X034 M143
3026 X035 M144
3028 X036 M145
3030 X037 M146
* Oil Chart
3032 M8013 [MOV K0 D606]
3038 [> D610 K720]
[MOV K0 D610]
[MOV K0 D604]
[MOV K0 D605]
[MOV K0 D606]
D606 A/S 3095 B/D 3033 3058
D610 A/S 3038 B/D 23 28 3043
D604 A/S 3069 B/D 33 3048
D605 A/S 3082 B/D 38 3053
D606 A/S 3095 B/D 3033 3058


3063 M200
INT D300 D607
ADD D604 D607 D604
3076 M201
INT D300 D607
ADD D605 D607 D605
3089 M202
INT D300 D607
ADD D606 D607 D606
* Charting
3102 T0
INT D296 D600
INT D64 D601
INT D130 D602
INT D76 D603
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
D607 A/S 3069 3082 3095 B/D 3064 3077 3090
D604 A/S 3069 B/D 33 3048
D607 A/S 3069 3082 3095 B/D 3064 3077 3090
D604 A/S 3069 B/D 33 3048
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
D607 A/S 3069 3082 3095 B/D 3064 3077 3090
D605 A/S 3082 B/D 38 3053
D607 A/S 3069 3082 3095 B/D 3064 3077 3090
D605 A/S 3082 B/D 38 3053
D300 A/S 2484 2522 2560 2693 2698 B/D 1249 1268 1287
D300 A/S 3064 3077 3090 B/D
D607 A/S 3069 3082 3095 B/D 3064 3077 3090
D606 A/S 3095 B/D 3033 3058
D607 A/S 3069 3082 3095 B/D 3064 3077 3090
D606 A/S 3095 B/D 3033 3058
D296 A/S 1229 3103 B/D 1216
D600 B/D 3103 B/D
D64 A/S 565 587 1164 1602 3108 B/D 369
D601 B/D 3108 B/D
D130 A/S 2433 2445 3113 B/D 837 856 936
D602 B/D 3113 B/D
D76 A/S 407 1111 1337 1425 3118 A/S
D603 B/D 3118 B/D


3123
[END]

Position	Device	Setting Value
(1)	T0	K10
(4)	T1	K30
(7)	T2	K10
(10)	T3	K50
(13)	T4	K110
(1467)	T5	D302
(2480)	T10	K1
(2495)	T11	K2
(2518)	T12	K1
(2533)	T13	K2
(2556)	T14	K1
(2571)	T15	K2
(2605)	T16	K2
(2638)	T16	K2
(2671)	T16	K2
(2790)	T100	D316
(2804)	T100	D316
(2818)	T100	D316

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
M0	*	*(1)		
M1	*	*(1)		
M2	*	*(1)		
M3	*	*(1)		
M4	*	*(1)		
M5	*	*(1)		
M6	*	*(1)		
M7	*	*(1)		
M8	*	*(1)		
M9	*	*(1)		
M10	*	*(1)		
M11	*	*(1)		
M12	*	*(1)		
M13	*	*(1)		
M17	*	*(1)		
M18	*	*(1)		
M19	*	*(1)		
M20	*	*(1)		
M21	*	*(1)		
M22	*	*(1)		
M23	*	*(1)		
M24	*	*(1)		
M25	*	*(1)		
M50	*	*(1)		
M100	*	*(4)		
M101	*	*(4)		
M102	*	*(4)		
M103	*	*(4)		
M104	*	*(4)		
M105	*	*(4)		
M106	*	*(4)		
M107	*	*(4)		
M108	*	*(4)		
M109	*	*(4)		
M117	*	*(2)		
M200	*	*(5)		
M201	*	*(5)		
M202	*	*(5)		
M203	*	*(2)		
M204	*	*(2)		
M205	*	*(2)		
M206	*	*(3)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
M304	*	*(1)		
M305	*	*(1)		
M306	*	*(1)		
D0	*	*(2)		
D1	*	*(1)		
D2	*	*(2)		
D12	*	*(1)		
D13	*	*(1)		
D20	*	*(2)		
D21	*	*(1)		
D22	*	*(1)		
D23	*	*(1)		
D42	*	*(1)		
D43	*	*(1)		
D44	*	*(2)		
D45	*	*(2)		
D64	*	*(1)		
D65	*	*(1)		
D100	*	*(1)		
D101	*	*(1)		
D102	*	*(1)		
D103	*	*(1)		
D104	*	*(1)		
D105	*	*(1)		
D106	*	*(1)		
D107	*	*(1)		
D108	*	*(1)		
D109	*	*(1)		
D110	*	*(1)		
D111	*	*(1)		
D112	*	*(1)		
D113	*	*(1)		
D114	*	*(2)		
D115	*	*(2)		
D116	*	*(2)		
D117	*	*(2)		
D118	*	*(2)		
D119	*	*(2)		
D120	*	*(2)		
D121	*	*(2)		
D122	*	*(1)		
D123	*	*(1)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
D130	*	*(3)		
D131	*	*(3)		
D132	*	*(1)		
D133	*	*(1)		
D134	*	*(1)		
D135	*	*(1)		
D136	*	*(1)		
D137	*	*(1)		
D138	*	*(1)		
D139	*	*(1)		
D200	*	*(1)		
D220	*	*(1)		
D221	*	*(1)		
D278	*	*(8)		
D280	*	*(1)		
D281	*	*(1)		
D282	*	*(1)		
D283	*	*(1)		
D284	*	*(1)		
D285	*	*(1)		
D286	*	*(1)		
D287	*	*(1)		
D288	*	*(1)		
D289	*	*(1)		
D290	*	*(1)		
D291	*	*(1)		
D292	*	*(1)		
D293	*	*(1)		
D294	*	*(1)		
D295	*	*(1)		
D296	*	*(1)		
D297	*	*(1)		
D298	*	*(3)		
D299	*	*(3)		
D300	*	*(4)		
D301	*	*(4)		
D302	*	*(3)		
D303	*	*(7)		
D304	*	*(4)		
D306	*	*(1)		
D307	*	*(1)		
D308	*	*(3)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
D309	*	*(3)		
D310	*	*(3)		
D311	*	*(3)		
D312	*	*(3)		
D313	*	*(3)		
D314	*	*(3)		
D315	*	*(3)		
D316	*	*(6)		
D317	*	*(2)		
D318	*	*(1)		
D319	*	*(1)		
D320	*	*(1)		
D321	*	*(1)		
D322	*	*(1)		
D323	*	*(1)		
D324	*	*(1)		
D325	*	*(1)		
D326	*	*(1)		
D327	*	*(1)		
D328	*	*(1)		
D329	*	*(1)		
D330	*	*(1)		
D331	*	*(1)		
D332	*	*(1)		
D333	*	*(1)		
D334	*	*(1)		
D335	*	*(1)		
D550	*	*(1)		
D604	*	*(3)		
D605	*	*(3)		
D606	*	*(3)		
D607	*	*(3)		
D610	*	*(3)		
T0	*	*(1)		
T1	*	*(1)		
T2	*	*(2)		
T3	*	*(2)		
T4	*	*(1)		
T5	*	*(1)		
T10	*	*(2)		
T11	*	*(2)		
T12	*	*(2)		

Data Name : Device List

Find In:Main_2
Find What:Used Device (Contact & Coil)
Print Range:Whole Range

*:in use (times): the number of coil uses

Device	Contact	Coil	Parameter	Comment
T14	*	*(2)		
T16	*	*(6)		
T100	*	*(3)		
Z0	*	*(3)		
Z1	*	*(3)		
Z2	*	*(3)		
Z3	*	*(4)		
Z4	*	*(3)		
Z5	*	*(3)		
P0	*	*(1)		
P1	*	*(1)		
P2	*	*(1)		
P3	*	*(1)		
P4	*	*(1)		
P5	*	*(1)		

[Line Statement List]

Step	Line Statement
0	Timers
22	Load Once
43	E-Stops
57	Automation
66	Declarations
134	Analog to Digital
168	ADC Float
310	Flowmeter to RPM
406	Level % Converter
615	Error Threshold
826	Water Temperature Target Logic
949	Heater latching
1100	Manual Heater Control
1110	Height to Mass
1163	Dynamic Upper Thermal Boundary
1239	PO Designation
1300	PO Timer (D302)
1464	PO Dispensing
1482	Tank Scoring
1600	Need Scheduler
1635	Scheduler Cases
1653	Mode Scheduler
1707	Manual Mode Scheduler
1717	Mode Indiacator
1741	LSM Scheduler
1907	HSM Scheduler
2213	FSM Scheduler
2387	Scheduler Gates
2431	Solar Gate
2456	Manual Solar Gate
2460	Oil Scheduler
2692	Oil Distribuition Calculation
2772	Oil Distribution Gating
2784	Oil Distribution Timing
2840	Jacket Pump
2845	AODD Pump
2878	Manual Pump Mode
2968	HMI Display for Inputs
3032	Oil Chart
3102	Charting

[P Statement List]

Device	P Statement
P0	RTD to Temperature
P1	Ultrasonic to Level
P2	Sensor Averaging
P4	Sensor Logging (5 sec)

Data Name : Project Contents List

Workspace Name :
Project Name : Main_2
Title :

Data Name	Last Change	Title
Parameter	9/6/2026 12:12:22 PM	
PLC Parameter	9/6/2026 12:12:22 PM	
Network Parameter	9/6/2026 12:12:22 PM	
CC-Link	9/6/2026 12:12:22 PM	
Global Device Comment	9/6/2026 12:12:23 PM	
Program Setting		
Execution Program		
MAIN	9/6/2026 12:12:23 PM	
MAIN	9/9/2026 1:05:29 PM	
POU		
Program	9/6/2026 12:12:23 PM	
MAIN	9/9/2026 1:05:29 PM	
Local Device Comment		
Device Memory	9/6/2026 12:12:37 PM	
MAIN	9/6/2026 12:12:37 PM	